



Medtronic

Stealth AXiS™ ENT Clinical Application User Manual

Instructions for ENT procedures using electromagnetic localization



Stealth AXIS™ version 1.0

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Intended use / Indications for use

The Stealth AXiS™ Surgical System, with the Stealth AXiS™ ENT clinical application, is intended for precise positioning of surgical instruments and as an aid for locating anatomical structures in open, minimally invasive, and percutaneous ENT procedures. Their use is indicated for medical conditions in which the use of stereotactic surgery may be appropriate, and where reference to a rigid anatomical structure, such as the skull, can be identified relative to images of the anatomy.

This can include, but is not limited to, the following procedures:

- Functional endoscopic sinus surgery (FESS)
- Endoscopic skull base procedures
- Lateral skull base procedures

Contraindications

None

Device description

The Stealth AXiS™ Surgical System is a computer-assisted surgery system that is composed of a platform, clinical application, surgical instruments, and a referencing system (which includes patient and instrument trackers). The system tracks the position of instruments in relation to the surgical anatomy, known as localization, and then identifies this position on preoperative or intraoperative images of a patient. The Stealth AXiS™ ENT clinical application helps guide surgeons during ENT procedures such as functional endoscopic sinus surgery (FESS), endoscopic skull base procedures, and lateral skull base procedures.

Patient images can be displayed by the ENT clinical application from a variety of perspectives (axial, sagittal, coronal, oblique) and 3-dimensional (3D) renderings of anatomical structures can also be displayed. During navigation, the system identifies the tip location and trajectory of the tracked instrument on images and models the user has selected to display. The surgeon may also create and store one or more surgical plan trajectories before surgery and simulate progression along these trajectories. During surgery, the clinical application displays how the actual instrument tip position and trajectory relate to the pre-surgical plan, helping to guide the surgeon along the planned trajectory. While the surgeon's judgment remains the ultimate authority, real-time positional information obtained through the Stealth AXiS™ Surgical System can serve to guide this judgment.

Stealth AXiS™ Core is composed of the workstation cart and the optional camera cart, which contain the key navigation components including the localizers and the computer with the operating system.

Compatibility and installation

The clinical application may be used on a Stealth AXiS™ planning station in an office environment, a Stealth AXiS™ computer-assisted surgery system in an operating room or office environment, or optionally from a web browser or a connected computer if the remote planning feature has been enabled on a Stealth AXiS™ planning station.

To use the remote planning feature, the connected computer must meet the minimum system performance requirements, operating system and browser compatibility requirements, and be able to access the planning server on the network. Configuring a planning station as a planning server requires the installation of a software license by qualified Medtronic personnel and further configuration by hospital staff. See the *Stealth AXiS™ IT Guide* for server and web browser configuration instructions.

The clinical application is compatible with Stealth AXiS™ systems and Medtronic navigational instruments.

The clinical application is installed by Medtronic service personnel only. If you have any questions about your clinical application installation, contact Medtronic Technical Services.

Note: When the Stealth AXiS™ clinical application is used with other medical devices, see the user instructions for those devices for additional labeling limitations.

Navigational accuracy

Under representative worst-case configuration, the Stealth AXiS™ Surgical System with the Stealth AXiS™ ENT clinical application has demonstrated performance in 3D positional accuracy with a mean error ≤ 2.0 mm and in trajectory angle accuracy with a mean error ≤ 2.0 degrees. The positional error is defined as the Euclidean distance from target to instrument tip at final point of insertion. The trajectory angle error is the angle between the planned surgical trajectory and trajectory of the instrument at final point of insertion. Depending on the surgical procedure and the particular instruments used, the resulting navigational accuracy may be negatively impacted.

This performance was determined using an anatomically representative phantom and utilizing a subset of system components and features that represent the worst-case combination of all potential system components in a clinically relevant workflow. The

test configuration included CT images with slice spacing and thickness of 1.25 mm and T2-weighted MR images with slice spacing and thickness of 1.0 mm.

User profile

The intended user profile for the system with the ENT clinical application is ENT surgeons, their support staff, hospital biomedical engineering support staff, medical imaging staff, and hospital IT.

Intended patient population

The intended patient population for the ENT clinical application is patients for whom stereotactic image-guided surgery is appropriate.

Side effects

The system and its associated applications are auxiliary surgical tools and do not independently produce side effects. When used in conjunction with navigated instruments for surgical procedures, the associated side effects are the same as those for the procedure itself.

Potential complications

The potential complications for Stealth AXiS™ Surgical System ENT Procedures identified by Medtronic include, but are not limited to:

General surgical complications, pain, insufficient treatment, infection, tissue or bone damage, burn, vision impairment, bleeding or blood loss, electric shock, cerebrospinal fluid leakage or fistula, brain or nervous system injury, death.

Clinical benefits

Navigated procedures performed using the Stealth AXiS™ Surgical System with the Stealth AXiS™ ENT clinical application potentially confer the following clinical benefits:

- Meta-analyses on commercial image-guided navigation systems suggest a reduction in major complications during endoscopic sinus surgery compared with non-image guided procedures.
- Image-guided navigation systems used in endoscopic sinus surgery have shown a reduction in the need for revision surgeries.
- Navigation improves visualization, enabling surgeons to work within complex sinus anatomy and optimize surgical strategies for their patients.

Warnings

Refer to the applicable package inserts and system manual for additional instructions, warnings, and cautions.

Warning: The system hardware and clinical application should be used only by qualified medical professionals who are trained in performing surgery and familiar with computer-assisted surgery systems.

Warning: Abort use of the system if navigation seems inaccurate and steps to restore accuracy are unsuccessful.

Warning: Inspect all visible system components before use, including the localizers that accompany your system (that is, the side emitter or the flat emitter). If visibly damaged, do not use the system. Abandon use of any component damaged during the procedure. Contact Medtronic technical services.

Warning: Inspect sterile device packaging and expiration date before use. Do not use any pre-sterilized device whose sterile packaging appears to be compromised or whose expiration date has passed. If sterile device packaging is compromised or damaged, contact Medtronic technical services.

Warning: Prior to use, examine instruments and blades for damage and deterioration. Do not use any bent, damaged, or compromised instruments or blades. Periodically inspect instruments or blades for damage during their use. Abandon use of any instruments or blades bent or damaged during the procedure. If instruments and blades are bent, damaged, or compromised, contact Medtronic technical services.

Warning: Do not reprocess, resterilize, or reuse any single-use device. Attempts to reprocess or resterilize a single-use device may be ineffective and may compromise its structural integrity. Any reuse creates a risk of contamination which could result in patient injury, illness, or death.

Warning: Disposable devices are provided in sterile condition for single use only. After use, dispose of single-use devices according to local and national regulations.

Warning: Device components may be supplied in a non-sterile condition and must be cleaned and sterilized prior to use. Verify that all necessary instrumentation has been properly cleaned and sterilized before surgery. For sterilizable components, refer to the cleaning and sterilization instructions that accompany the component. For non-sterilizable components, refer to the *Stealth AXiS™ Surgical System Cleaning and Disinfection Instructions*.

Warning: Do not open sterile-barrier packages or containers until surgical use.

Warning: Do not modify the Stealth AXiS™ Surgical System, components or accessories, without prior written consent from Medtronic Navigation. Prohibited modifications include but are not limited to: altering, repairing, or replacing system components; altering software provided by Medtronic; updating the operating system or drivers; and installing any software that is not provided by Medtronic. Secure the system to prevent unauthorized modifications. Unauthorized modifications could render the system unsafe or ineffective for its intended use.

Warning: Use care during handling, storage, and cleaning of instruments to prevent physical injury and deformation of the instrument tip.

Warning: Do not allow fluid to enter the electrical components of tracked instruments.

Warning: Do not reuse or reprocess a device where transmissible spongiform encephalopathy agents or Creutzfeldt-Jakob disease contamination is suspected or confirmed. Incinerate the potentially contaminated device according to national guidelines.

Warning: To prevent cross-contamination, always handle, transport, and reprocess devices that contact the central nervous system separately from other devices.

Warning: To avoid potential exposure to blood-borne pathogens and chemicals, use appropriate Personal Protective Equipment when handling or disposing of Medtronic devices.

Warning: The side emitter and the flat emitter are not intended for sterile patient contact or interaction. If you place these items in the sterile field, follow proper aseptic technique guidelines to bag or drape them prior to use.

Warning: Do not use the side emitter or the flat emitter in ambient (room) temperatures greater than 30°C (86°F). Exceeding this temperature limit could result in navigational inaccuracy or patient or user harm from excessive temperatures.

Warning: RF emitters typically found in an operating room can affect medical electrical equipment performance, such as the Stealth AXiS™ electromagnetic localization system.

Warning: Before transporting the system carts, shut down the system and stow all components, dock the monitors, dock the camera on the camera cart, close all storage compartments, and remove any loose items from the top of the carts.

Warning: The system and its associated applications contain no user-repairable parts. For repair or replacement of any part of the system or application, contact Medtronic technical services.

Precautions

Caution: Federal law (U.S.A.) restricts this device to sale by or on the order of a physician.

Caution: The system mouse, side emitter, flat emitter, and emitter holder are not designed for sterilization and may be damaged if sterilization is attempted.

Caution: Metallic or conductive objects in or near the electromagnetic navigation field can degrade navigational accuracy and the navigation volume. If metallic distortion causes excessive error, navigation may be disabled. To restore navigation, move metallic and conductive objects away from the navigation field.

Caution: Electrical noise in or near the electromagnetic navigation field can degrade navigational accuracy and the navigation volume. If electrical noise introduces excessive error, the system may automatically disable navigation. To restore navigation, move devices that produce electrical noise (such as electrocautery equipment and electric drills) away from the navigation field.

Caution: For procedures performed with the flat emitter, a surgical table with a radiolucent table top is recommended. If the surgical table has a metal frame or metal side rails, position the flat emitter at least 5 cm (2 in) away from the metal. Positioning the emitter closer may result in reduced navigational accuracy and volume.

Caution: If the surgical table to be used with the side emitter has a metal frame or metal side rails, position the side emitter at least 25 cm (10 in) away from the metal. Positioning the emitter closer may result in reduced navigational accuracy and volume.

Caution: Procedures performed with the side emitter require a surgical bed featuring side rails because the arm assembly that supports the emitter connects to the side rails.

Caution: The system is not suitable for use in the presence of a flammable, anesthetic mixture with air, oxygen, or nitrous oxide.

Contact information

If a serious incident occurs in relation to the use of the system, report it to Medtronic Navigation. If a serious incident occurs in the European Union, also report it to the competent authority in the Member State where the incident occurred.

For instructions about return shipping or about transporting system components or accessories, contact Medtronic technical support.

Telephone	Technical support worldwide	+1 720 890 3160
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Interacting with the clinical application: Touchscreen monitor and mouse

The touchscreen monitor is the primary interface for the system; therefore, these instructions include monitor interactions such as tap, double-tap, and drag. Instructions to tap an object on the screen mean to touch the monitor with a finger, a gloved finger, or a capacitive stylus and then lift the finger or stylus quickly.

You can also use a mouse to interact with the clinical application. When using a mouse, instructions to “tap” an object on the screen are equivalent to click on the object with the left mouse button.

For a list of touchscreen interactions and their equivalent mouse actions, see “Touch and mouse interactions.”

Document conventions used in this manual

This document employs the following conventions:

- Warnings are preceded by the word **Warning**. Failure to observe a warning may result in serious injury, death, or other serious adverse reactions to the patient or operator. Pay special attention to these items.
- Precautions are preceded by the word **Caution**. Failure to observe a precaution could result in injury, damaged equipment, forfeited time or effort, or the need to abort use of the system.
- References to buttons that appear on the touchscreen monitor are enclosed in square brackets. For example: Tap **[Edit...]**.
- References to buttons that are icons without text are capitalized. For example: Tap the Expand button...
- References to menu options that appear on the touchscreen monitor are printed in bold letters. For example: Select **Clear** from the list.
- References to titles of specific instructions for use documents are in italics.
- Electronic instructions for use can be accessed in two different ways: at www.medtronic.com/manuals and within the clinical application. To view the instructions for use in the clinical application, tap **[Help]**, and tap **[Go to list of instructions for use]**.

Planning options

You can create and modify surgical plans in the **Planning** task by using the workstation cart, the Planning Station, or in a remote planning session. For more information about the Remote Planning feature, see “Remote planning.” For more information about the Planning Station, see the *Stealth AXiS™ Planning Station System Manual*.

Remote planning

The Remote Planning feature allows you to plan procedures both inside and outside the operating room, provided your computer has hospital network access and the facility has set up a Planning Station server. Using a supported web browser, you can view patient data and create surgical plans in the clinical application. No patient data is stored on your computer. Plans are saved to the Planning Station server and can later be imported to a Stealth AXiS™ workstation for use in the operating room.

While a remote planning session is active, the selected patient case is locked to prevent simultaneous edits by multiple users. If another user selects the same surgeon profile, the procedure is also locked. Users are encouraged to log in with their own credentials or create their own unique profile.

Browser sessions time out after a period of inactivity, with a warning message displayed beforehand. The timeout duration is set by hospital IT staff and can last between 10 and 60 minutes.

Warning: Review patient data created using remote planning before making surgical decisions and actions.

Note: Images may take a longer time to load when using the Remote Planning feature.

Note: In a remote planning session, you cannot use a CD/DVD or USB drive to import or export images or surgeon profiles.

Minimum laptop or desktop requirements

To use remote planning, you only need to install the Google Chrome™ browser on your laptop or desktop—no additional client or software installation is required. Refer to a Windows or Apple user manual or technical support to confirm what your hardware and software specifications are.

Note: If you don't comply with the following recommendations, remote planning might not work on your laptop or desktop.

Note: If your network connection is poor, confirm that you have both a stable and fast internet connection. Consider moving your desktop or laptop to a new location.

Minimum computer screen and resolution recommendations:

- Minimum screen size: 13 inches
- The following screen resolutions have been tested:
 - 2560 x 1440
 - 1920 x 1080
 - 1440 x 810

Minimum Windows hardware and software recommendations:

- Windows 11 (64-bit) operating system
- Intel® Core™ i5* 10th generation or later processor, or equivalent AMD processor
- Memory: 16 GB RAM
- Graphics:
 - Integrated Graphics Card: Intel® Arc™* graphics card or equivalent AMD graphics card
 - Dedicated Graphics Card: NVIDIA RTX™ A1000* graphics card or equivalent AMD graphics card

Minimum Apple hardware and software recommendations

- Operating System: macOS Sonoma 14, macOS Sequoia 15
- Processor: Apple M3 or Apple M1 Pro
- Memory: 8 GB RAM
- Graphics: Apple M3 or Apple M1 Pro

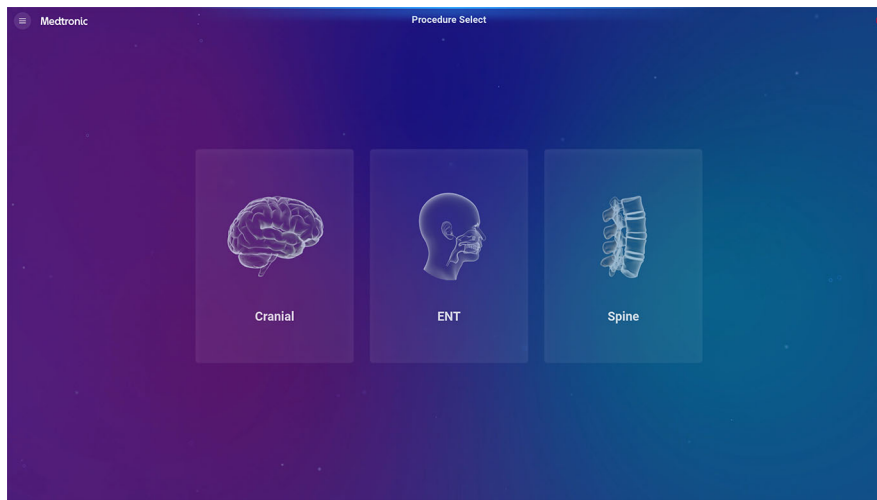
Guidance for optimizing Windows laptops or desktops

The following guidelines are recommendations for Windows laptops or desktops running Windows 11 equipped with two GPUs. Visualization performance is optimized with remote planning by using the dedicated GPU instead of the integrated GPU.

1. Confirm that the Chrome™ browser is installed on your laptop.
 - a. Press the Windows key on your keyboard, type **Graphics settings**, and then press "<Enter>."
 - b. Click **[Add desktop app]**.
 - c. Click in the address bar, type **C:\Program Files\Google\Chrome\Application** and press "<Enter>."
 - d. Click **chrome.exe** and then **[Add]**. Depending on your system settings, you may not see the .exe suffix. It may only be labeled "chrome".
2. Select **Google Chrome** and click **[Options]**.
3. Select the dedicated GPU that you want to utilize.
4. Close all Chrome™ windows and restart the Chrome™ browser for the changes to take effect.

Logging in to the system from a cart

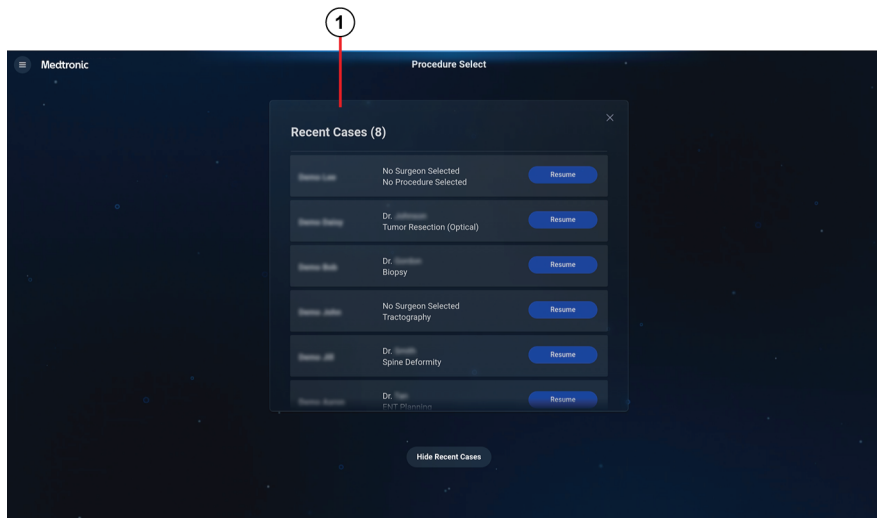
1. At the login screen, enter your username and password.
 - Tap **[Log In]**.
 - If you are unable to log in to the system, contact your hospital's Stealth Admin or IT department to confirm login information. Medtronic technical services can provide a temporary passcode for emergency system access. Tap **[EMERGENCY AND SERVICE LOGIN]** and call Medtronic technical services for assistance.
2. To launch the clinical application, tap the desired clinical application icon or tap **[View Recent Cases]**.



Resuming recent cases

1. Tap **[View Recent Cases]** in the **Procedure Select** screen to view a list of patients imported or selected from the last 36 hours.
2. Tap **[Resume]** to select the desired patient case.

The **Images** task displays for the last surgeon and procedure used with this patient.



1 Recent Cases

Selecting a surgeon and procedure

After selecting the desired clinical application icon, the **Procedure Select** screen displays.

Follow these steps to select a surgeon and a procedure.

1. Tap **[Standard]** or select a surgeon name.

Alternately, tap **[Add Profile]** to add a surgeon profile in the **Preferences** task. If a procedure is locked by another user, you can create a copy of the procedure.

Surgeon profiles allow you to save surgeon-specific preferences such as instruments, navigation options, and view settings. These preferences are organized by procedure. For each surgeon profile, you can also control the display of the on-screen keyboard and the colorblind aid.

2. After selecting a surgeon profile, select the surgical procedure to be performed.

Alternately, tap **[Add Procedure]** to create or customize procedures in the **Preferences** task. See “Customizing procedures” for more details.

Images

Importing images

On a Stealth AXIS™ system, you can import patient images from multiple sources (for example, CD/DVD or USB media, other Stealth AXIS™ systems, or a hospital PACS). Images must meet certain requirements. For information on scan requirements, see the *Stealth AXIS™ Imaging Protocols for Navigated Procedures*.

Tap the History button to show or hide a list of entries for each previous user session. The history list includes username, time stamp, and certain actions such as plans created. This may be helpful to detect unexpected edits to the plan by someone else. The History button is disabled if the patient is currently selected by another user on a remote planning session.

You can import DICOM (Digital Imaging and Communications in Medicine) patient images in the **Images** task. Tap a tab at the top of the screen:

- Use the **Import** tab to import images from a PACS, another Stealth AXIS™ system, CD/DVD or USB media, or to import Demo Exams.
- Use the **Patients On This Stealth** tab to select images that are already on the system.

To import NIfTI (Neuroimaging Informatics Technology Initiative) images, see “Importing NIfTI images.”

Note: A Stealth AXIS™ archive or a StealthStation™ S8 application v2.1 or later archive can also be imported.

Note: To transfer an unregistered scan from the O-arm™ system to the Stealth AXIS™ system, refer to the user manual that accompanies the O-arm™ Imaging System.

Importing images from a network source (DICOM)

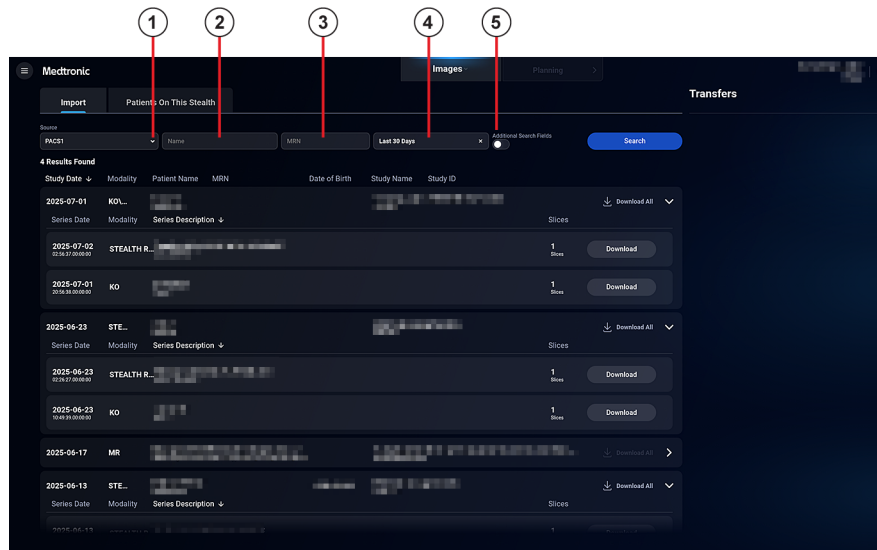
Note: If an image has been manually pushed from a network source (DICOM) to the Stealth AXIS™ system, it will automatically appear under the **Patients On This Stealth** tab.

1. Tap the **Import** tab, if it is not already selected.
2. Select a network source from the **Source** drop-down menu.
If you need to add a network source, see “DICOM Setup.”
3. Enter search criteria.
 - a. Enter information in any combination of the **Name** and **MRN** (Medical Record Number) fields. The patient name search is not case sensitive.
 - b. Tap the date filter field (**Last 30 Days** is the default filter) to display date search options (select **Last 24 Hours**, **Last 7 Days**, **Last 30 Days**, **All Time**, or a custom date range).
 - c. Turn on the **Additional Search Fields** toggle to show the **Referring Physician** field and the **All Modalities** toggle.
 - d. Turn off the **All Modalities** toggle to filter by MR, CT, or Stealth modalities.

Note: To see all data from the network source, leave the search fields blank.

Note: Refine search criteria for faster searches. Leaving all search fields blank may result in long search times.

4. Tap **[Search]**.



- 1 **Source** drop-down menu
- 2 **Name** filter field
- 3 **MRN** filter field
- 4 Date filter field (**Last 30 Days** is the default)
- 5 **Additional Search Fields** toggle (turns on and off additional search filters)

5. Import the images needed for the procedure. The first study in the list is expanded by default. Tap other studies in the list to display associated series.
 - To import an individual series, tap **[Download]** beside the series.
 - To import all series for the study, tap **[Download All]**.

A list of patients displays in the **Transfers** list after a download is complete. The **Active Transfers** provides a download status for each image series.

6. After downloading all necessary images, proceed to preview the images.

Importing images from media

On a Stealth AXIS™ system, you can import patient images located on CD/DVD or USB media. This option is not available for remote planning.

1. In the **Images** task, tap the **Import** tab.
2. Insert the CD/DVD or USB media that contains DICOM images into the system.
3. If more than one media type is available, or media was already inserted before tapping the **Import** tab, select the desired media from the **Source** drop-down menu and tap **[Search]**.
4. In the **Select a directory to search** dialog, search the folders to find the desired images.

Note: The clinical application searches for DICOM images in subfolders, but manually selecting a lower level folder will result in faster search times.

5. Tap **[OK]**.
The clinical application searches for DICOM images in the selected folder.
6. Select a patient from the search results list.
7. Import the images needed for the procedure.
 - To import an individual series, tap **[Download]** beside the series.
 - To import all series for the selected study, tap **[Download All]**.
8. After downloading all necessary images, proceed to preview the images.

Importing NIfTI images

On a Stealth AXIS™ system, you can import NIfTI images located on CD/DVD or USB media. This feature is not available for remote planning.

1. In the **Images** task, tap the **Import** tab.
2. Insert the CD/DVD or USB media that contains NIfTI images into the system.
3. If more than one media type is available, or media was already inserted before tapping the **Import** tab, select the desired media from the NIfTI section of the **Source** drop-down menu and tap **[Search]**.
4. Browse the media file system and select a folder that contains NIfTI images.
5. Tap **[OK]**.
The clinical application searches for NIfTI files in the selected folder.
6. When the search results appear, select a file.
Note: You can tap **[Select All]** to select all the available files.
7. You can import NIfTI files as new patients or existing patients.
 - Tap **[Create New Patient]** and enter a **Name** and **MRN** to import the files to a new patient.
 - Tap **[Select Existing Patient]** and select a patient from the drop-down menu to import the file to a patient who is already in the system.
8. Tap **[Import]**.
9. After downloading all necessary images, proceed to preview the images.

Viewing patient status and history

If your hospital facility is running a planning server, a lock icon next to a patient case indicates that another user has the patient case selected. If so, the system prevents you from selecting the same patient. From the **Patients On This Stealth** tab you can view the patient status on the patient card. The name of the user who has the patient currently selected displays and the patient is not selectable.

If a procedure is locked, a second user might be using the same surgeon profile to look at that procedure. For this reason, it is recommended that you create your own unique profile that is not used by others.

If you want to make a copy of a patient case while it is currently in use by someone else, you can clone the patient. See "Cloning patients" for more information.

Tap the History button to show or hide a list of entries for each previous user session. The history list includes username, time stamp, and certain actions such as plans created. This may be helpful to detect unexpected edits to the plan by someone else. The History button is disabled if the patient is currently selected by another user on the system.

Previewing images

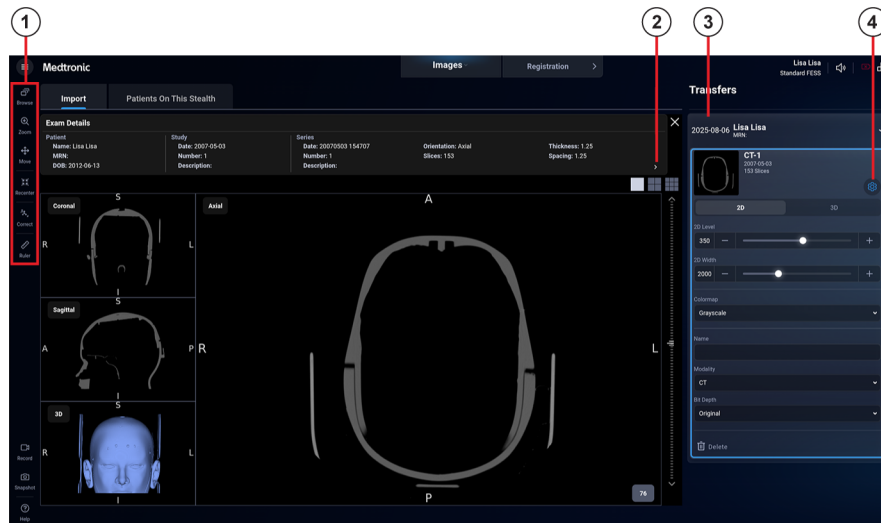
When your patient's name appears in the list of imported images:

1. Tap the patient's name. The clinical application displays all of the series associated with the patient in the **Patient Images** list.
Note: If images are not under the same patient, see "Managing patients and images" for instructions on how to combine patients or move images.
2. Tap on the series that you plan to use for navigation. The clinical application displays preview images and image details.
Options to browse and view additional slices include:
 - Tap and drag the slice indicator on the vertical scroll bar or use the mouse wheel.
 - Tap above or below the slice indicator to move one slice in the indicated direction.
 - Tap the arrow at the top or bottom of the scroll bar to move one slice in the indicated direction.
3. In **Exam Details**, tap the Expand button to view more image details.
4. Preview the images.
 - a. Make sure that the anatomy of interest is captured in the images.
 - b. Make sure that the images do not contain geometric distortion or intensity distortion (intense brightness in parts of the image compared to the rest of the image). If images contain distortion, contact radiology.
 - c. Check the slice spacing and slice thickness. Series with slice spacing equal to slice thickness are recommended.

- d. To change the view orientation of a selected procedure, see “Customizing procedures.” If you have not selected a procedure, you can change the view and patient orientation while in this task.
 5. If the image orientation is incorrect in the scan, you can correct it:

Warning: Only correct the image orientation if it is incorrect in the scan. Verify that all aspects of the patient images are correct, including patient data and image orientation. Make sure that the image orientation matches the patient’s anatomy. Incorrect image orientation can result in inaccurate navigation and loss of data.

 - a. To rotate the image, tap **[Correct]** and tap the desired icon in the **Rotate** tab. Tap **[Confirm]**.
 - b. To reorient the image, tap **[Correct]** and tap the **Reorient** tab. Then tap the desired orientation button. Tap **[Confirm]**.
 6. If you need to adjust the level or width, tap the Settings button on the series in the **Patient Images** list. Then tap the **2D** or **3D** tab and use the **Level** and **Width** sliders.
- For additional information about the buttons on this screen, see “Buttons in Images, Planning, or Navigation.”
- Merging multiple series from the same patient requires the ENT StealthMerge™ license. See “Using image merge” for more details.
7. If you plan to merge multiple series from the same patient:
 - a. Tap each series in the **Patient Images** list to preview the images and check the image quality.
 - b. Turn on the **Select** toggle in an image series to add the images to the merge set.
 8. After confirming that everything is correct in the patient's images, tap the next desired task at the top of the screen.



- 1 Tools for modifying images
- 2 The Expand button displays additional image details
- 3 Patient images
- 4 The Settings button shows image display options such as level and width adjustments

Planning

In the **Planning** task and in the **Navigation** task, you can use a variety of tools to define surgical plans to target structures appropriate for your procedure.

After you define a surgical plan, you can change the plan in the following ways:

- Edit the target point
- Edit the entry point
- Edit a waypoint
- Change the plan name
- Change the plan color
- Lock the plan to prevent inadvertent changes
- Show or hide the plan

See “Editing a surgical plan” for instructions on editing plans.

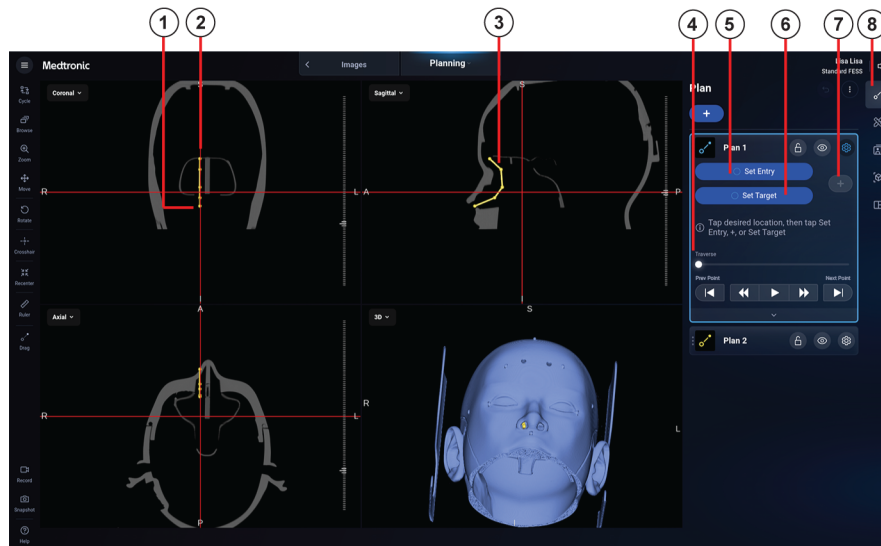
Warning: Verify that a plan’s target point, waypoints, and entry point are clinically valid before using it for a procedure. Traverse the plan to check the entry, waypoints, and target after any edit. If you change the merge reference image after the plan is created, review your surgical plan to avoid inaccurate navigation. Some existing plan data is remerged to the new reference image and may have shifted.

Defining a surgical plan

To define a surgical plan:

1. Tap the Plans button to open the **Plans** drawer.
2. Scroll through the 2D image slices and find the slice that contains the surgical target.
3. Tap a target point in one of the 2D views. Refine its location by tapping in any of the other views.
4. Tap **[Set Target]**.
5. Tap a plan Settings button to open the plan details.
6. Tap an entry point on one of the 2D views or on the surface of the 3D model.
7. Tap **[Set Entry]**.
8. If you want to create a waypoint, tap an intermediate location on one of the 2D views. Tap **[+]** to create a waypoint.
9. Create additional waypoints between the entry and target as needed for the surgical plan.
10. Simulate movement along the plan by doing one of the following:
 - Drag the **Traverse** slider.
 - Tap the Backward and Forward buttons to move along the plan by -1.0 mm or +1.0 mm.
 - Tap Next Point and Prev Point to move along the plan toward the entry or target location, including waypoints that are added in between.
 - Use the left, right, up, and down arrow keys on the keyboard.
 - Tap the Play button to automatically traverse the plan.

If you move a point by mistake, tap the Undo button to undo one change.
11. If you want to create another plan, tap **[+]** in the **Plans** drawer and repeat the steps above.



- 1 Plan entry
- 2 Plan target
- 3 Plan trajectory
- 4 **Traverse** slider
- 5 **[Set Entry]**
- 6 **[Set Target]**
- 7 Set Waypoint
- 8 Plans button

Editing a surgical plan

- Use the touchscreen to edit a target, entry, or waypoint by moving the crosshairs and tapping the Settings button, then tapping **[Set Point]**.
- Tap the Drag button to drag the entry or target and rotate the plan in any 2D view.
- While holding the <SHIFT> key, you can use the mouse to drag the entry, target, or waypoint and rotate the plan in any 2D view.

Editing the target

1. Select the plan you want to edit from the **Plans** drawer.
2. Tap the Skip Forward button in the drawer until you reach the target.
3. In any view, tap the location of the desired target.
To magnify the entire image, tap **[Zoom]**, and then drag with one finger up to zoom in or down to zoom out.
4. Tap the Settings button, and then tap **[Set Point]**.
The clinical application sets the target to the point indicated by the crosshairs.

Editing a waypoint

1. Tap the plan you want to edit from the **Plans** drawer.
2. Tap the waypoint that you want to edit in the drawer.
3. In any view, tap the location of the desired waypoint.
To magnify the entire image, tap **[Zoom]**, and then drag with one finger up to zoom in or down to zoom out.
4. Tap the Settings button on the waypoint you want to edit, and then tap **[Set Point]**.
The clinical application sets the waypoint to the point indicated by the crosshairs.

Editing the entry

1. Select the plan you want to edit from the **Plans** drawer.
2. Tap the Skip Backward button in the drawer until you reach the entry for the plan.
3. In any view, tap the location of the desired entry.
To magnify the entire image, tap **[Zoom]**, and then drag with one finger up to zoom in or down to zoom out.
4. Tap the Settings button, and then tap **[Set Point]**.
The clinical application sets the entry to the point indicated by the crosshairs.

Locking plans

Follow these steps to lock a plan to prevent changes.

1. In the **Plan** drawer, tap the plan you want to lock.
2. Tap the Lock button to lock the selected plan.
3. To unlock a plan, tap the Lock button again.

Changing plan names and colors

1. Select the plan you want to edit from the **Plan** drawer.
2. To change the plan name, tap the Settings button, and then tap the Expand button to enter text in the **Name** field.
3. To change the plan color, tap the Expand button, and then select one of the predefined colors, or double-tap a predefined color and select a custom color.

Deleting a surgical plan

To delete a plan do one of the following:

- Tap the Settings button, tap the Expand button, and then tap **[Delete]**.
- Swipe right and then tap the Delete button.
- Tap the More button and then tap **[Delete Multiple]** or **[Delete All]**.

Equipment

Setting up and managing equipment

The **Equipment** task displays system components and external devices used in the procedure. You can manage the equipment for the procedure and confirm that the equipment is communicating with the system.

To enter the **Equipment** task, tap the Menu button in the upper left corner of the screen and then tap **Equipment**.

The **Add Equipment** list contains the equipment available for the procedure.

- To add equipment to the procedure, tap the Add button.
- Some equipment requires additional information before you can add it to the procedure. Complete the necessary fields.

The equipment status bar and connection lines display communication status. Solid green lines show verified communication, and dashed orange lines show lack of communication.

- To remove equipment from the procedure, tap the Delete button.
- Tap the Settings button on the equipment card to:
 - display information for a part of the system that is not communicating.
 - display information about the equipment.
 - tap some equipment fields to edit them.

When all the necessary equipment is communicating, proceed to the next task in the clinical application.

Attaching a patient reference

To navigate the patient's anatomy, you must attach a reference frame to the patient's head.

Three patient reference frame options are available with the ENT clinical application:

- The Non-Invasive Patient Tracker (NIPT) (9734887, 9734887XOM)
- The Skull-Mount Patient Tracker (SMPT) (9735300, 9731132)
- The EM ENT Head Frame (9733445, 9733445XOM)

Warning: Do not use the patient reference frame as a landmark for confirming navigational accuracy. The position of the anatomy is defined by the position of the patient reference frame. If the patient reference frame moves with respect to the anatomy after you register the patient, navigation becomes inaccurate. Make sure that the patient reference frame is secure. If the frame moves in relation to the patient anatomy after registration, you must register the patient again before navigating.

Option 1: Non-Invasive Patient Tracker (NIPT)

(Minimally-invasive procedures only)

Note: The Non-Invasive Patient Tracker has a divot and arrows for use in the ENT clinical application.

1. Position the patient on the surgical table and rest the patient's head on a gel headrest or similar support.
2. Review the warnings and other instructions given in the package insert that accompanies the Non-Invasive Patient Tracker (9734887, 9734887XOM).
3. Select a site on the patient's skin to affix the tracker.

Warning: Carefully consider the location where you will place the Non-Invasive Patient Tracker. Choose a skin site that fulfills these criteria:

- Will lie safely within the localizer detection field during navigation.
- Will not result in stress on the tracker as a result of patient positioning.
- Will not hinder access to the surgical site.
- Will not be in the immediate vicinity of any large metal objects.
- The skin is taut and smooth, not loose, baggy, or uneven. The tracker's position relative to the operative anatomy may not be allowed to change during the procedure. The patient's forehead is the recommended application site for the Non-Invasive Patient Tracker.

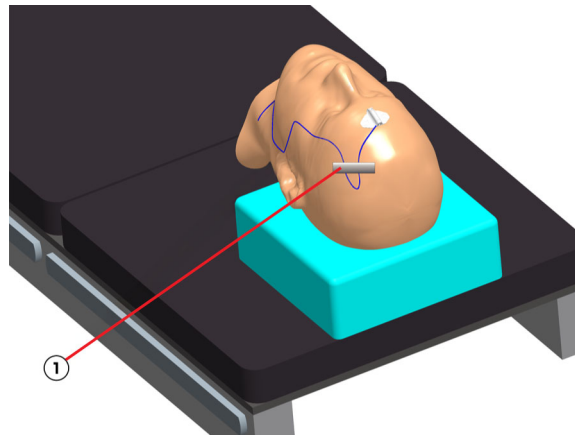
Warning: The Non-Invasive Patient Tracker may not exhibit the same accuracy as reference frames using rigid fixation.

4. If necessary, shave the site to provide maximum contact with the patient's skin.
5. Clean the site on the patient's skin with alcohol to strengthen adhesion and allow the site to dry thoroughly before applying the tracker.

Note: If the patient is perspiring, you may want to apply benzoin or a similar product to improve adhesion.

6. Peel away the release liner from the tracker and adhere the tracker to the patient's skin.
7. Create a U-shaped bend in the tracker's cable and tape this bend to the patient approximately 5 cm (2 in) below the tracker. This creates a strain relief mechanism that helps prevent the tracker from being dislodged accidentally.
8. Connect the patient tracker communication cable to any open port on the EM instrument interface.
9. Verify that the tracker is secure and cannot be accidentally dislodged during the procedure.

Warning: Do not bump or reposition the patient reference frame after registration. Movement of the patient reference frame relative to the patient anatomy will result in inaccurate navigation. If the patient reference frame moves in relation to the patient anatomy at any time after registration, you must re-register.



- 1 Tape the cable to the patient to create a strain relief.

Option 2: Skull-Mount Patient Tracker (SMPT)

This option is recommended for procedures requiring a large skin flap, such as tumor resections.

Warning: The safe use of the SMPT has not been demonstrated with pediatric patients.

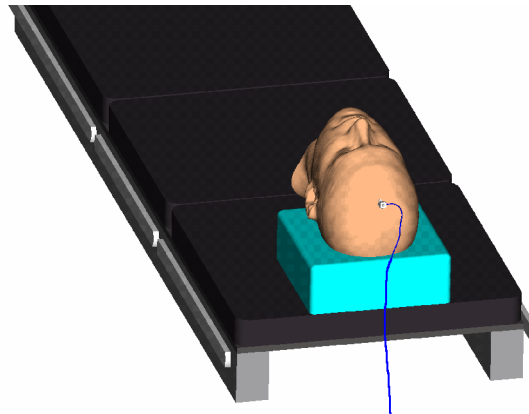
Warning: Before attaching the SMPT carefully examine the patient image data to verify that the skull thickness at the intended attachment site is adequate for securing the SMPT and screws, and that no arterial or venous structures lie beneath the site. When fully tightened, the SMPT screws penetrate 2.5–3.0 mm below its lower surface. Never place the SMPT at a site where the skull bone is less than 4.0 mm thick.

1. Position the patient on the surgical table and rest the patient's head on a gel headrest or similar support.
2. Review the other warnings and instructions given in the package insert that accompanies the sterile, disposable SMPT (9735300, 9731132).
3. Select and prepare the attachment site.
4. Make a small scalp incision (14–16 mm) down to the bone at the attachment site.
5. Set the sterile SMPT at the site and tighten the screws using the sterile screwdriver (reusable or disposable). Because the screws are self-tapping and self-drilling, no pilot holes are necessary.

Caution: Make sure that the screws are secure so the patient tracker cannot slip or shift at any time during the procedure. However, be careful not to over-tighten the screws.

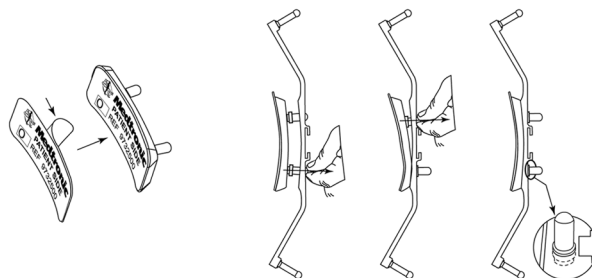
6. Connect the patient tracker communication cable to any open port on the EM instrument interface.

Warning: Do not bump or reposition the patient reference frame after registration. Movement of the patient reference frame relative to the patient anatomy will result in inaccurate navigation. If the patient reference frame moves in relation to the patient anatomy at any time after registration, you must re-register.

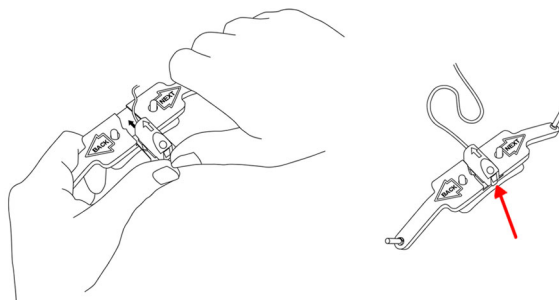


Option 3: Assembling and attaching the head frame

1. Remove the paper covering the back of the adhesive pad and attach the pad to the bottom of the silicone pad.
2. Attach the silicone pad to the head frame.



3. Ensure that the head frame is free from debris and then attach the EM ENT Patient Tracker to the head frame. Slide the tracker into the slot until the stop snaps against the frame (red arrow). Make sure that the cable extends over the patient's head (not over the face).



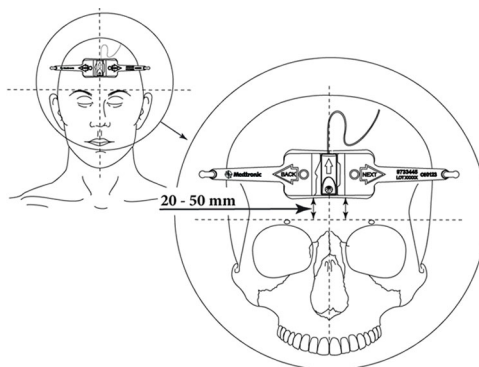
Warning: Inspect the tracker and the cord for damage, nicks, or exposed wire. Never use a damaged tracker.

4. Remove the paper from the front of the adhesive pad.
5. Clean the mounting area on the patient's forehead with an alcohol wipe. Allow the area to dry.

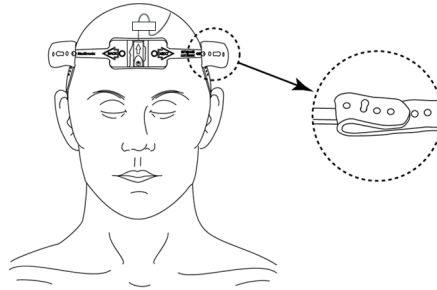
Warning: Always screen the patient for adverse reaction to all patient contacting materials, especially adhesives, prior to the procedure.

Warning: Do not mount the head frame to irritated or wounded skin; skin reaction may occur.

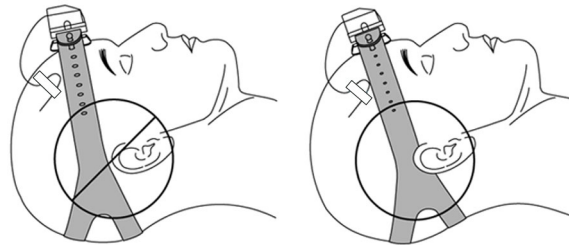
6. Place the head frame on the center of the forehead 20–50 mm (1–2 in) above the supra orbital foramen nerve.



7. Pull the strap up to the arm pins and hook the strap onto the pins with light, even tension. Make sure that the strap is centered laterally and that the extra strap length is roughly equal on the left and right sides. The bifurcated part of the strap should be as close to the base of the skull as possible.



8. Increase the tension on the strap by working back and forth from right to left one hole at a time until the frame is stable and resistant to slippage. Make sure that the tension is even across the strap, to prevent lateral drift during surgery.



Warning: Do not over-tighten the strap. Look for signs of excessive pressure surrounding the EM ENT Head Frame site. Over-tightened straps may cause pressure-related injuries such as bruising, skin marks, or nerve damage.

Warning: To avoid inaccuracies in navigation position, ensure that the head frame placement does not compress the skin toward the brow.

9. Push gently on the head frame to verify that it is securely attached and cannot shift with respect to the head at any time during the procedure. If it slips during this test, remove it, and reattach it.
10. Secure any excess strap length back over the pins.

Warning: If it becomes necessary to replace the EM ENT Patient Tracker, remove the EM ENT Head Frame from the patient first. If you remove or replace the EM ENT Patient Tracker, you must re-register the patient. To remove the EM ENT Patient Tracker from the EM ENT Head Frame, place both thumbs on the cable side of the tracker and push it out of the slot.

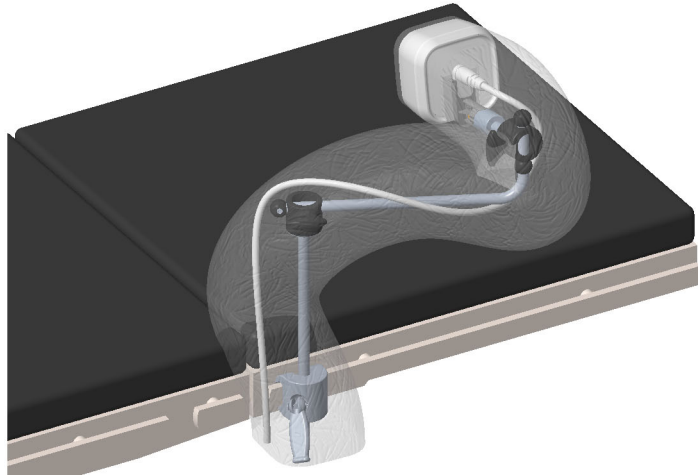
Note: Always replace the adhesive pad if it becomes necessary to replace the head frame during the procedure. Make sure that the area is clean, dry, and free of oil and hair.

11. Connect the EM ENT Patient Tracker cable to any open port on the EM instrument interface.
12. Create a U-shaped bend in the EM ENT Patient Tracker cable and tape the bend to the patient's forehead. This creates a strain relief mechanism that helps prevent the EM ENT Patient Tracker from being accidentally dislodged.

Draping the side emitter

Warning: The side emitter and the arm assembly that supports the side emitter are not intended for sterile contact or patient interaction. If you place these items in or near the sterile field, bag or drape them. Failure to drape these accessories can compromise the sterile field.

1. Drape the side emitter, articulating arm, and rails clamp radial.
2. If you are using the Medtronic drape (9736701), cuff the drape using the tabs.



3. Leave some space in the drape around the locking knob to allow for repositioning.

Setting up and positioning the side emitter

Warning: Visually inspect all accessory devices before use. Do not use accessory devices if they are visibly damaged.

Warning: Do not place the side emitter in contact with the patient because the emitter temperature may reach 48°C (118°F).

Caution: Procedures performed with the side emitter require a surgical bed featuring side rails because the arm assembly that supports the emitter connects to the side rails.

Note: To reduce electromagnetic tracking distortions, use titanium or plastic retractors (not stainless steel).

1. Connect the side emitter cable to the emitter port on the workstation cart localization I/O panel.
2. Set up the side emitter where it will not interfere with the surgeon's access to the patient during surgery.

Setting up the emitter holder

1. Connect the emitter support bracket (9735547) to the FlexENT™ articulating arm (9736175).
2. Attach the rails clamp radial (5.750-2366) to the surgical table or to the ENT clamp rail on the workstation cart by sliding the rails clamp radial over the end of the mounting rail.
3. Insert the articulating arm into the clamp and tighten the handle.



- 1 Rails clamp radial (5.750-2366)
- 2 FlexENT™ articulating arm (9736175)
- 3 Emitter support bracket (9735547)
- 4 Mounting location (surgical table or workstation cart ENT clamp rail)

4. Loosen the two ball-joint tightening knobs and position the articulating arm.

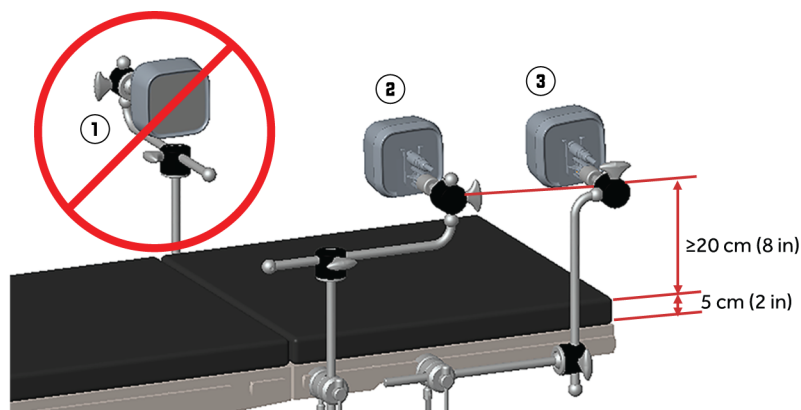
Caution: Always loosen the knobs before adjusting the articulating arm.

Caution: Moving parts may create pinch points. Use care to maintain control of the object you are moving and be aware of potential pinch points when moving articulating arm joints.

Caution: When adjusting the articulating arm, remove or hold on to the side emitter before loosening the adjustment knob. Dropping the side emitter could void its calibration.

Note: Exact placement of the emitter support bracket depends on many factors including the type of table, procedure, patient location, and patient size. Use a configuration that suits your situation. For best results, follow these practices:

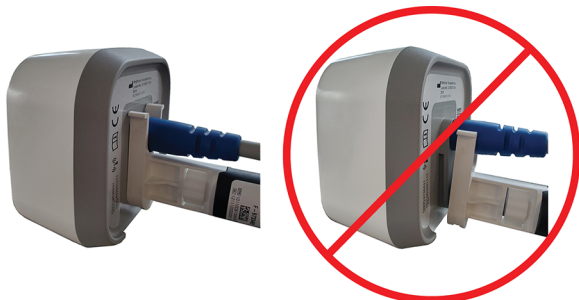
- a. Orient the ball joints as far away from the emitter's electromagnetic field as possible.
- b. Make sure that no part of the articulating arm extends into the electromagnetic field.
- c. Position the emitter support bracket at least 25 cm (10 in) above the metal on the surgical table and point it toward the trackers.



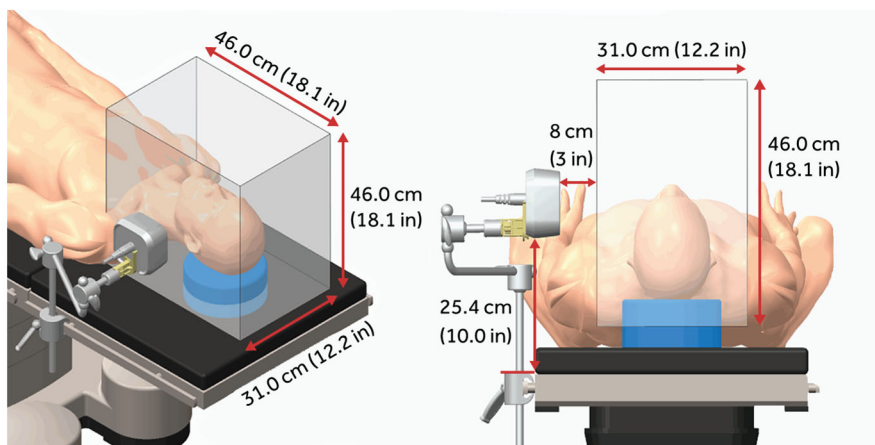
- 1 Example of incorrect setup (articulating arm extends into the electromagnetic field)
- 2 Example of correct setup
- 3 Example of correct setup

Draping the side emitter

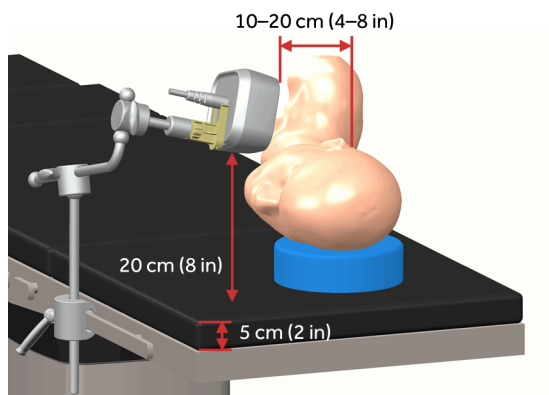
5. Tighten the two ball joint knobs.
6. If using the workstation cart, move the cart into position and lock the caster brakes.
7. Place the back side of the side emitter flush against the emitter support bracket with the logo on the emitter oriented up and slide the emitter into the bracket until the emitter snaps into place.



Note: For FESS and Endoscopic Skull Base procedures, the emitter is most effective when it points directly at the patient's nose and is between 15 and 25 cm (6 and 10 in) away from the nose.



Note: For Lateral Skull Base procedures, the emitter is most effective when it points at the patient's ear and is between 10 and 20 cm (4 and 8 in) away from the ear.



Refining emitter position

Before setting up for sterile navigation, check the geometry error metric in the clinical application. For best results, minimize the geometry error metric by adjusting the emitter's position or by positioning the patient closer to the emitter.

1. Connect an instrument to any port on the EM instrument interface.
2. In the **Instruments**, **Registration**, or **Navigation** task, tap the Emitter button to open the tracking view, if it is not already open.
3. Tap **[Open Tracking Details]** to display the tracking details.
4. Confirm that the emitter and EM instrument interface are connected to the system.
5. Make sure that the clinical application can track the instrument and the patient reference frame throughout the range of locations required for registration and navigation.
6. Adjust the position of the emitter as necessary until the instrument and patient reference frame display in the tracking view.

For the flat emitter, if your volume is smaller than expected, decrease the padding thickness between the patient and the emitter or add padding between the emitter and the metal on the surgical table. Gradually increase the separation distance between the emitter and the metal by a centimeter or more while maintaining proper patient positioning, and re-evaluate the volume.

Draping the flat emitter

Warning: The flat emitter is not intended for sterile contact or interaction. If you place the flat emitter in the sterile field, bag or drape it.

Drape the flat emitter before placing the patient on the emitter.

1. Place a drape over the flat emitter.
2. Make sure that the drape completely covers the flat emitter.

Note: For patient comfort, you can place a sterile head cushion on the center of the flat emitter. Make sure that the cushion is centered on the etched crosshairs on the flat emitter. A sterile gel pad can also be placed below the flat emitter for shoulder positioning.

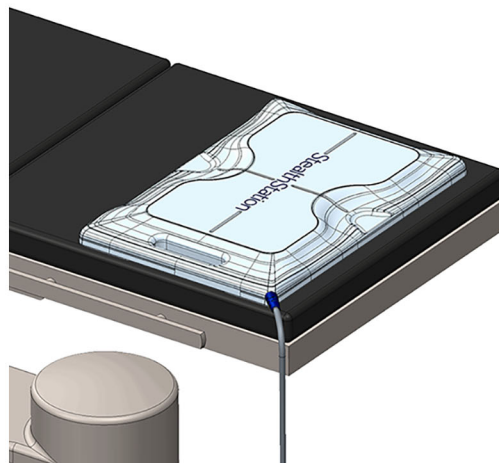
Setting up the flat emitter

Caution: Do not exceed the maximum weight load of 6.5 kg (14 lbs) on the flat emitter. Exceeding the maximum weight load may cause damage to the emitter.

Connect the flat emitter cable to the emitter port on the workstation cart localization I/O panel.

1. Place the flat emitter at the head of the surgical table and position it at least 5 cm (2 in) above the metal of the surgical table with the logo oriented up.

Note: Do not place the flat emitter directly on the metal of the surgical table, and do not place the flat emitter over the pedestal of the surgical table.

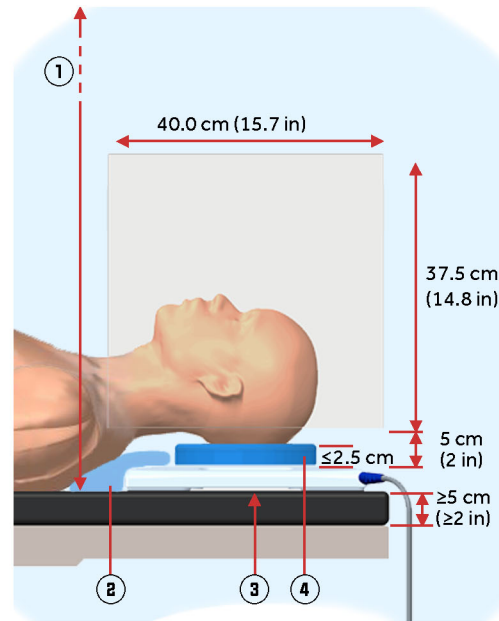


Draping the flat emitter

2. Make sure that the flat emitter cable rests off to the side of the surgical table and does not interfere with the surgeon's access to the patient during surgery.

Note: Make sure that there is no tension on the flat emitter cable.

3. For best results, place the surgical area of interest near the center of the navigation field.

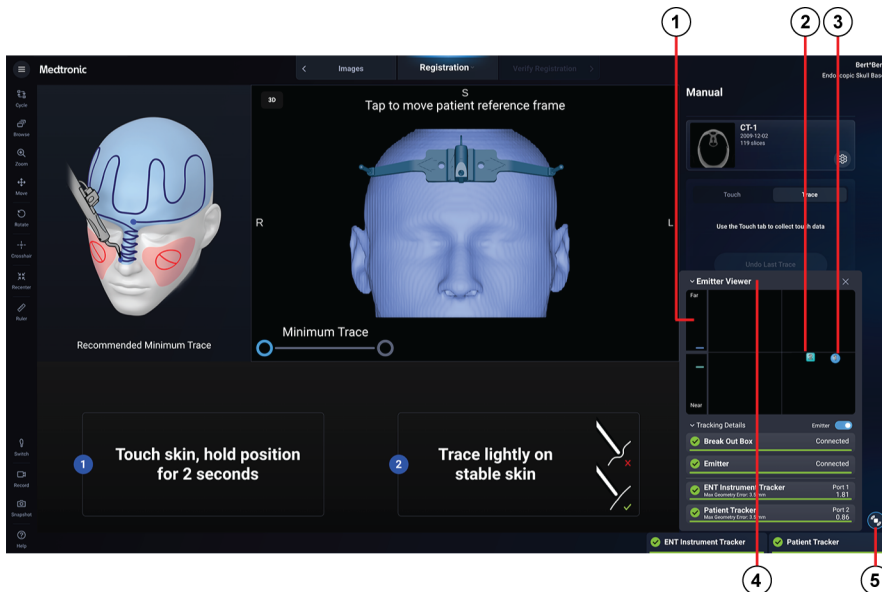


- 1 Low-energy magnetic field: Metal objects, such as surgical lights, in this area will reduce the size of the navigation field and impact navigational accuracy.
- 2 Towel for patient comfort
- 3 Location to place additional padding, if necessary, to increase the separation distance between metal rails on the table and the flat emitter
- 4 Thin pad: 2.5 cm (1 in) or less

Tracking view

The tracking view shows the location of tracked instruments and the patient reference frame in the navigation field. The distance indicator shows the instrument's and patient reference frame's distance from the localizer.

- To open the tracking view, tap the instrument status bar, the patient reference frame status bar, or the Emitter button.
- For more detailed information about the navigated instrument and patient reference frame, tap **[Tracking Details]**.



- 1 Distance indicator
- 2 Patient reference frame
- 3 Instrument
- 4 **Emitter Viewer**
- 5 Optical display

Instruments

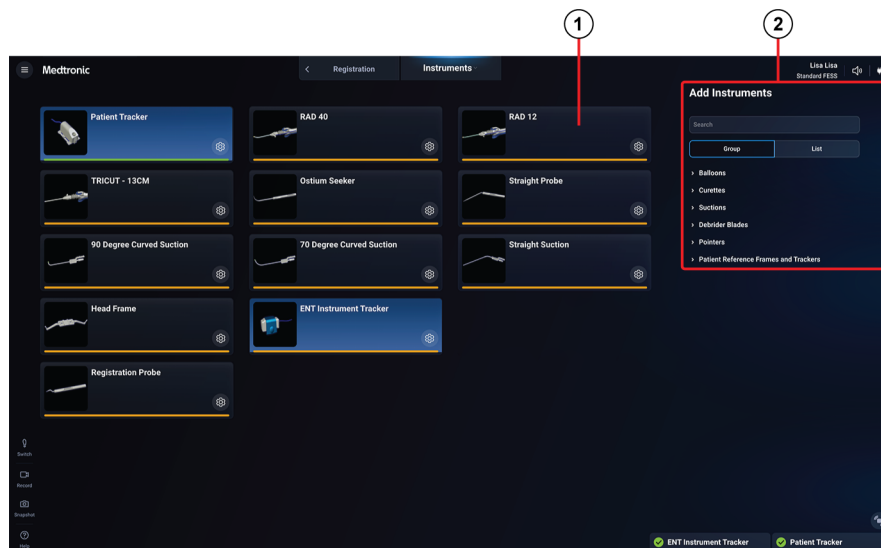
Instrument management

In the **Instruments** task, you can choose the appropriate instruments for your procedure and verify instruments.

The **Instruments** task displays tool cards for the instruments added to the procedure.

The **Add Instruments** section contains a list of instruments that you can add to your procedure. You can drag with one finger up or down on the touch screen or scroll through the list using a mouse wheel.

Note: Any EM trackers or instruments that are plugged into the EM instrument interface will appear automatically.

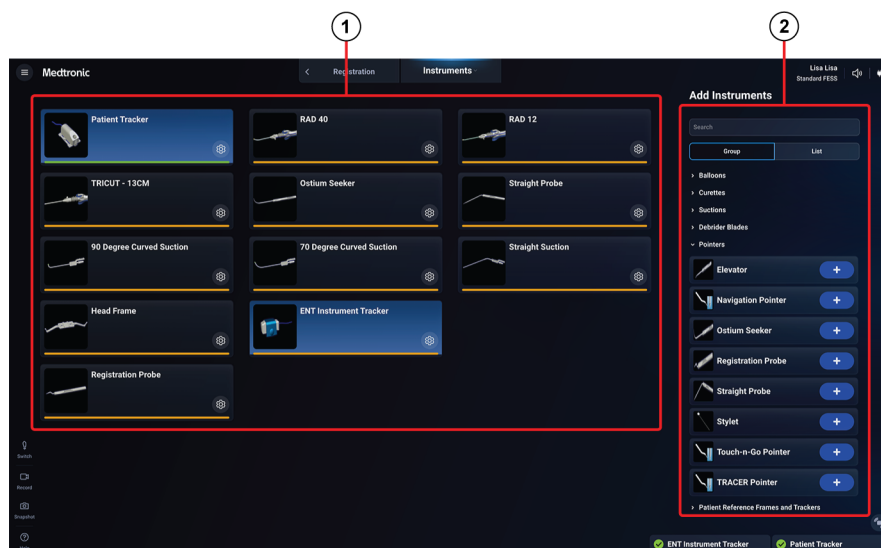


- 1 Instrument tool card
- 2 Add Instruments

Adding or removing instruments

You can add or remove instruments from your procedure in the **Instruments** task. To edit the instrument set for your procedure, follow these steps:

1. Plug ENT trackers and EM instruments into the EM instrument interface. Instruments that are plugged in will appear in the procedure automatically.
2. Attach an ENT instrument tip to the ENT tracker and verify the instrument. ENT instrument tips will appear in the procedure when they are verified.
3. In the **Add Instruments** menu, scroll through the list of available instruments or enter the name of an instrument in the search bar.



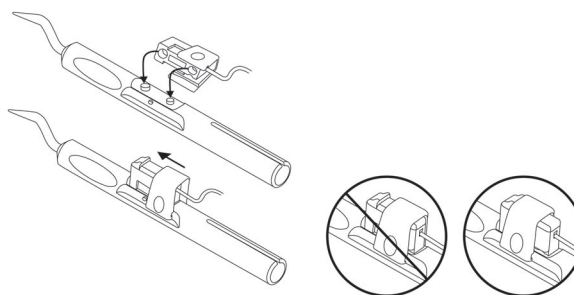
- 1 Instruments in Procedure
- 2 Add Instruments menu

Attaching instrument trackers

Warning: Inspect instruments for any defects. Inspect the tracker and the cord for damage, nicks, or exposed wire. Never attempt to use a bent or damaged instrument or tracker.

1. Make sure that the instrument handle is free of debris.
2. Place the EM ENT Instrument Tracker over the two posts on the instrument's handle. The tracker cable should exit towards the hind of the instrument.
3. Slide the clip toward the instrument tip. The clip is properly latched when it snaps in place against the stop.
4. Connect the EM ENT Instrument Tracker cable to a port on the EM instrument interface.

Note: When an EM ENT Instrument Tracker is within one inch of a metal object (for example, the endoscope's camera), the instrument may have red status and navigation may halt because the metal interferes with detection of the tracker. To achieve green status, move the tracker farther away from the metal object.



Detaching instrument trackers

1. Slide the clip toward the hind of the instrument.
2. Lift the EM ENT Instrument Tracker off of the two posts on the instrument's handle.

Verifying instruments with EM ENT trackers

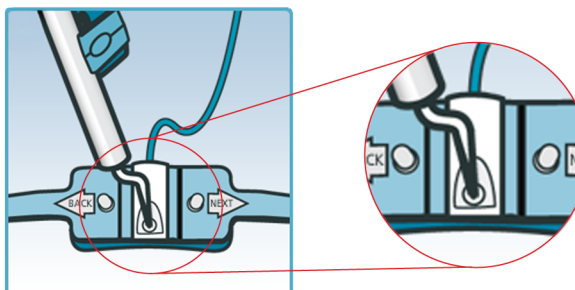
Warning: Always verify an instrument before using it. If you transfer an instrument tracker from one instrument to another, you must verify the instrument that you moved the tracker to.

Verifying Instruments ensures that the navigation system knows the location of the instrument's tip, and prevents the use of bent or damaged instruments.

You can verify instruments in the **Instruments**, **Registration**, and **Navigation** tasks.

To verify an instrument, follow these steps:

1. Make sure that the screen shows the correct patient reference frame.
2. Insert the tip of the instrument into the bottom of the divot in the patient reference frame. Keep the instrument as perpendicular to the divot as possible.



3. Tap the Tracking View button to open the tracking view and then confirm that both the instrument and the patient reference are in the emitter's field of view.
4. Hold the instrument steady until you hear a chime, or press the footswitch.

When an instrument is verified, the system makes a positive sound (chime). When an instrument verification fails, the system makes a negative sound (bonk). See "Troubleshooting instruments" for instrument verification troubleshooting tips.

5. If you switched the instrument tracker to a new instrument, the new instrument name will display upon successful verification. Before navigating, confirm that the system displays the correct name of the instrument. If the instrument displays red status, the instrument is outside the emitter's navigation field.

Using automatically verified electromagnetic (EM) instruments

Warning: Verify that the navigation system displays the name of the tracked tool and patient tracker.

Warning: Physically inspect instruments for any defects. Never attempt to use a bent or damaged instrument.

Medtronic EM instruments with integrated sensors are automatically verified. When you want to use an automatically verified instrument, follow these steps:

1. Plug the instrument and patient reference frame into open ports on the EM instrument interface.
2. Confirm that the clinical application displays the name of the tracked tool and patient reference frame.
3. Place the instrument and patient reference frame in the navigation field.
4. Confirm that both the instrument and the patient reference frame are visible in the tracking view.
5. Repeat this process for each instrument that you plan to use in the procedure.

Note: It is unnecessary to put the tip of automatically verified EM instruments into the patient reference frame divot.

Note: Plugging automatically verified EM instruments into the EM interface adds them to the active procedure.

Registration

Registration overview

To navigate, you must match the patient's image to the patient's anatomy using a process called registration. For ENT procedures, you use manual registration methods (trace and touch registration).

For manual registration, you may use touch registration, trace registration, or a combination of both. Both manual registration methods require a registration instrument such as the EM ENT Registration Probe.

For the trace and touch registration methods, only use image modalities containing anatomic information. The clinical application automatically creates a three dimensional model using the patient's images. Before you begin a manual registration, verify that the 3D skin model accurately represents the patient and that all skin fiducial markers are clearly visible if using touch registration. If the 3D model contains artifacts or scatter, edit the model to make it as accurate as possible (see "Editing a 3D registration model").

Note: If you edit the 3D model after you begin a registration, you will need to restart the registration process.

- In trace registration, you move the tip of the probe over regions of the skin that can be seen within the 3D skin model. For trace registration instructions, see "Performing trace registration."

Warning: Do not use trace registration for Lateral Skull Base procedures. Trace registration is not a registration option when using the default Lateral Skull base procedure provided by the clinical application.

- In touch registration, you place the tip of the probe on specifically defined landmarks on the patient's skin. You can also add discrete points on the skin's surface that are not near a landmark. For touch registration instructions, see "Performing touch registration."

Warning: Do not bump or reposition the patient reference frame after registration. Movement of the patient reference frame relative to the patient anatomy will result in inaccurate navigation. If the patient reference frame moves in relation to the patient anatomy at any time after registration, you must re-register.

If you merged images and want to change the image used for registration, you can change the image before starting manual registration. If you change images after registration, you will lose your current registration and must restart the registration. To change the image, tap the Settings button, and then tap **[Change Image]**.

Touch and Trace registration methods

The system uses all of the information that you enter on the **Touch** and **Trace** tabs to compute a single registration. You can use any combination of the touch and trace registration methods that fit your preferred work flow.

Follow these best practices when registering:

- Provide the system with as many registration points as you can reasonably include, but avoid tracing the same area repeatedly. Include points over a wide area that surround the surgical region of interest.
- Use a combination of touch and trace points if you are unable to surround the surgical region of interest using a single registration method.

- Ensure that the registration probe and patient reference frame are visible in the tracking view.
- Ensure that all the points you store on both the **Touch** and **Trace** tabs are accurate.
 - Check for inaccurate touch points and tracing.
 - Delete or undo any touch points or traces that you know are inaccurate. For traces, you can undo only the portion that you traced after the last batch of points was collected.
- After registration, verify the registration across multiple known anatomical landmarks. The **Registration Accuracy** value is an indication of the amount of error in your registration. For more information about **Registration Accuracy**, see “Registration accuracy.”

Trace registration

The following trace registration options are available for Standard FESS and endoscopic skull base procedures. You can enable or disable these options on the **Navigation** tab in your surgeon profile. See “Editing a procedure.”

- To initialize manual registration using the patient reference frame, see “Adjusting the patient reference frame position on the 3D model.”
- To orient manual registration using 3-point touch then initialize by tracing on the patient, see “Performing 3-point touch to orient before tracing.”
- To orient and initialize manual registration by tracing on the patient, see “Performing trace registration.”

Adjusting the patient reference frame position on the 3D model

In Standard FESS and endoscopic skull base procedures, you have the option to initialize registration by adjusting the patient reference frame position on the 3D model to match the location on the patient's anatomy. You can enable or disable this option with the **Patient Reference Frame** setting on the **Navigation** tab in your surgeon profile. See “Editing a procedure.”

If the option is enabled and you are using the Non-Invasive Patient Tracker (NIPT) or the EM ENT Head Frame, a representation of the patient reference frame appears on the 3D model. On the screen, move the patient reference frame on the 3D model so that its location matches the physical frame's location on the patient. Follow these steps to move the patient reference frame on the screen:

1. In the **Registration** task, tap the **Trace** tab.

Note: You can only move the frame on the 3D model on the **Trace** tab.
2. Look at the patient's anatomy to determine the location of the patient reference frame.
3. Tap the 3D model where you want the patient reference frame centered. The frame moves to the new location.

Note: You cannot move the frame on the 3D model after you begin tracing or if you have already performed a touch registration. If you need to move the frame after you have started tracing, tap **[Restart]** and then move the frame.
4. Proceed to “Performing trace registration.”

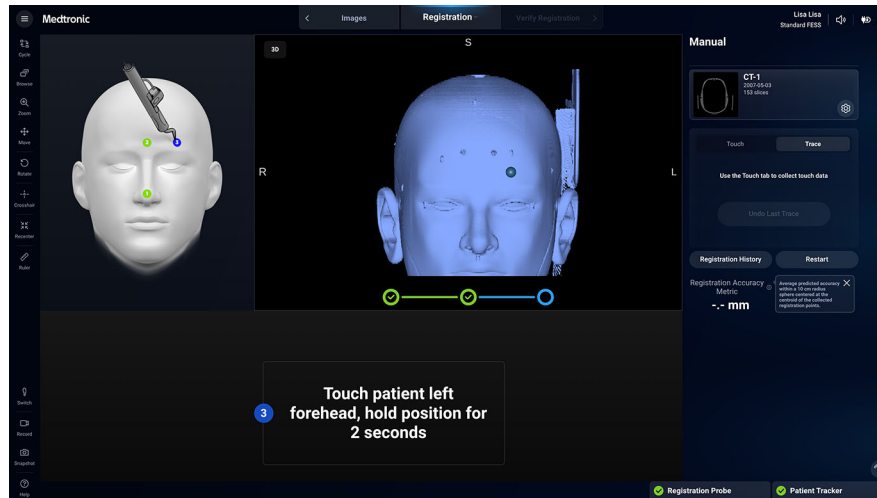
Performing 3-point touch to orient before tracing

Note: You have the option to use 3-point touch to orient the registration before you begin tracing. You can enable or disable the 3-point touch registration option using the **Patient Reference Frame** setting on the **Navigation** tab in your surgeon profile. See “Editing a procedure.”

Note: Depending on the **Registration Footswitch Behavior** setting, you can collect registration points by holding the registration probe steady for 2 seconds on a touched point or by pressing the footswitch.

Follow these steps to perform 3-point touch before tracing:

1. A blue dot will appear in the 3D model view. If necessary, adjust the position of the blue dot to the most anterior position on the patient (tip of the patient's nose). You cannot adjust any other blue dots after this step.
2. Touch the tip of the registration probe to the tip of the patient's nose, press and release the footswitch or hold the instrument steady for 2 seconds until you hear a verification sound.
3. Touch the second point on the patient's skin (center of the forehead or between the eyebrows), press and release the footswitch or hold the instrument steady for 2 seconds until you hear a verification sound.
4. Touch the third point on the left side of the patient's forehead, press and release the footswitch or hold the instrument steady for 2 seconds until you hear a verification sound.
5. Proceed to “Performing trace registration.”



Performing trace registration

Follow these best practices when performing trace registration:

- Create the full trace as a series of small trace segments by starting and stopping movement of the registration probe on the patient's anatomy. Each time you stop tracing, the clinical application adds the new segment to the registration and updates the **Registration Accuracy** value.
- If the last segment of the trace is inaccurate, delete it by tapping **[Undo Last Trace]**. You can remove up to the last 4 segments by tapping **[Undo Last Trace]**. If you need to remove more than 4 stored segments you must restart the registration.
- The **Registration Accuracy** value is an indication of the quality of the registration. Disregard small changes in the **Registration Accuracy** value between traces.
- When the registration is acceptable, verify the registration by touching known anatomical landmarks. See "Verifying registration."

Follow these steps to perform a trace registration:

1. In the **Registration** task, tap the **Trace** tab.
2. To begin trace registration, place the tip of the registration probe on the patient's anatomy and press and hold the footswitch. Move the tip of the probe lightly around the firm and bony areas of the patient's nose, brow, mastoids, and scalp to collect segments for the registration.

Note: If the **Stationary probe** setting is enabled for this procedure on the **Navigation** tab in your surgeon profile, you can start and stop collecting trace segments by holding the probe steady on the patient's skin. See "Editing a procedure." While tracing, the system emits an audio signal when the tracing state changes. One beep indicates the start of tracing. Two beeps indicate that tracing has stopped. One high-pitched beep indicates that initialization is complete. When initialization is complete, the progress bar prompts you to verify the registration or continue collecting trace segments.

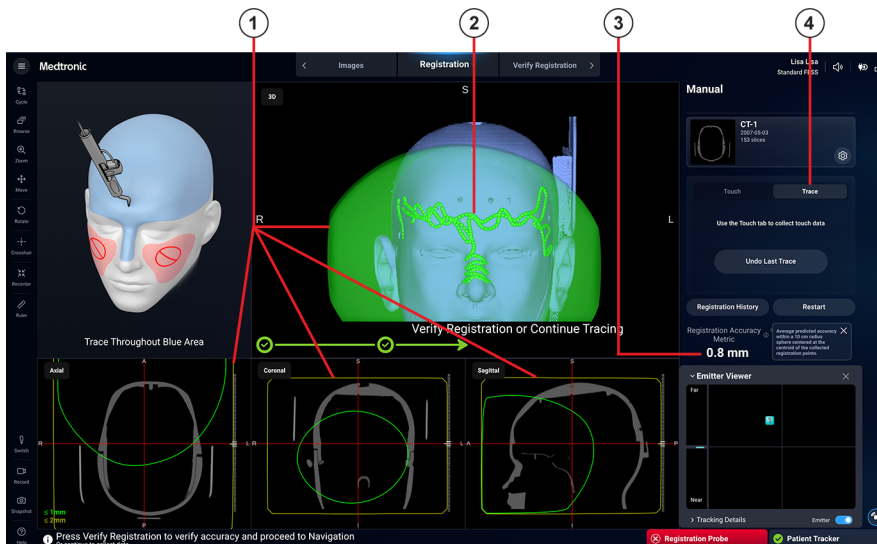
Follow these best practices for tracing:

- Gently maintain contact with skin at all times while tracing. If you need to lift the registration probe to move around an obstruction, release the footswitch (or hold the probe steady for 2 seconds if the **Stationary probe** setting is enabled) before you lift the probe. Then reposition the probe before you press the footswitch, or hold the probe steady for 2 seconds, to start tracing again.
- Include as many uniquely-shaped areas as possible, but only trace anatomy that is included in the patient image.
- Make sure that you collect segments both around the target anatomy and on the opposite side of the head.
- Do not collect segments in areas where the tissue is deformable or in areas such as the cheeks or neck where the skin may be loose.

When the **Minimum Trace** progress bar is full, the clinical application displays the points on the 3D model and displays the **Registration Accuracy** value. Lower numbers indicate a more accurate registration.

Note: If you started registration with 3-point touch or the patient reference frame, your trace segments will display blue until you reach the minimum amount of collected segments needed for trace registration.

For more information about the **Registration Accuracy** value and the Zone of Predicted Registration Accuracy, see “Registration accuracy.”



- 1 Zone of Predicted Registration Accuracy
- 2 Collected trace segments
- 3 Registration Accuracy
- 4 Trace tab

3. Review your trace registration.

- If you are satisfied with the Zone of Predicted Registration Accuracy and the **Registration Accuracy** value, tap **[Verify Registration]** to advance to the registration verification task.
- If you are not satisfied with the Zone of Predicted Registration Accuracy and the **Registration Accuracy** value, try the following to refine the trace:
 - Attempt additional tracing until you have collected segments over a broad area. Each time you stop tracing, the clinical application adds the new segment to the registration and updates the Zone of Predicted Registration Accuracy and the **Registration Accuracy** value.
 - If additional tracing does not produce a usable registration, tap **Touch** and then define and touch additional landmarks. See “Performing touch registration.” After touching the registration landmarks, tap **Trace** and continue tracing.
 - If the combination of trace and touch registration does not produce a usable registration, tap **[Restart]**.

If the **Registration Accuracy** exceeds 5.0 mm, the clinical application does not allow you to advance to **Navigation**. In that case, trace over a broader area or, if your trace contains many yellow or red segments, tap **[Restart]** to start the registration process again. Previous registrations are saved to **Registration History**.

Performing touch registration

Defining registration landmarks

The clinical application automatically detects IZI skin fiducial markers in the images and stores them as touch registration landmarks. You can also store anatomical landmarks for use in touch registration. For more guidance on fiducial placement, see *Stealth AXiS™ Imaging Protocols for Navigated Procedures*.

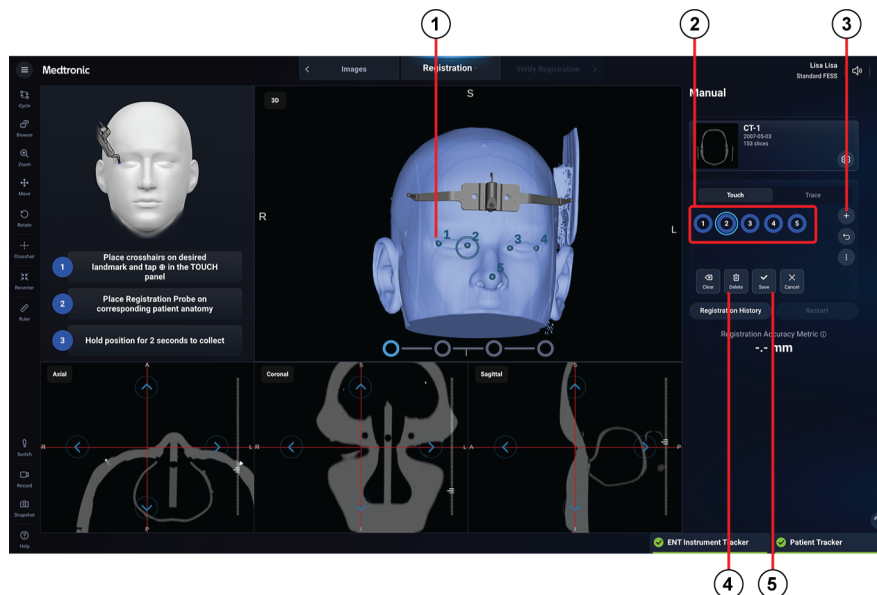
To store anatomical landmarks or undetected fiducial markers, follow these steps:

1. In the **Registration** task, tap the **Touch** tab.
2. Tap an identifiable anatomical landmark on the 3D model in the clinical application.

Suggested landmarks:

- Left tragus or right tragus
- Medial and lateral canthi

- Columella
 - Nasion
 - Tip of the nose
3. Tap **[+]**.
The clinical application adds a marker at that location on the 3D model and a touch registration point in the landmarks table.
 4. After a touch registration point is added, if necessary, refine the position of the selected point in the 2D views using the directional arrows and tap **[Save]**.
Note: You can cycle between two 2D image view layouts in the **Registration** task: **Trajectory 1 / Trajectory 2 / Probe's eye** and **Axial / Sagittal / Coronal**. If you tap **[Cycle]** to change the image view layout in the **Registration** task, the layout you select remains in the **Registration** task even if you exit and re-enter the procedure.
 5. Repeat these steps until you have selected up to 12 well-distributed landmarks. At least 4 landmarks are required to calculate a touch registration.
 - Select an asymmetrical set of landmarks.
 - Select landmarks that vary in distance from one another.
 - Select a set of landmarks that define a 3D space, rather than a straight line or a 2D plane.
 - Select anatomical or fiducial characteristics that are identifiable in both the clinical application image and on the patient (such as moles).
 - Select a set of points near the surgical area of interest.
 - Note:** If you initialized the registration using trace registration, there is no minimum number of touch points.
 6. You can use optional controls to manage the set of stored landmarks.
 - To delete a landmark, select it and tap **[Delete]**.
 - To assign a number to a detected fiducial marker, select its landmark and tap **[Assign]**.
- Note:** If a landmark was added manually, then a number will be automatically assigned.



- 1 Landmark
- 2 Landmarks table
- 3 **[+]**
- 4 **[Delete]**
- 5 **[Save]**

Editing registration landmarks

Check the locations of all stored landmarks to make sure that the locations are correct. Follow these steps to edit landmarks if necessary.

Note: Fiducial landmarks should be stored on the surface of the patient's skin in the center of the fiducial marker.

1. Tap a stored landmark in the **Touch** tab or on the 3D model.
2. Stretch a 2D view to zoom in to the precise location of the stored landmark.
3. Use the following methods to move the stored landmark to the correct location.
 - Tap a point in the 2D or 3D views. (Tapping near a stored landmark or detected fiducial marker in the 3D views snaps the cursor to the landmark or detected fiducial marker location.)
 - To refine the landmark position, use the directional arrows in the 2D views.
4. When the landmark is in the desired location, tap **[Save]**. Alternatively, tap **[Cancel]** to restore the landmark to its original location.

Touching landmarks

Warning: Do not bump or reposition the patient reference frame after registration. Movement of the patient reference frame relative to the patient anatomy will result in inaccurate navigation. If the patient reference frame moves in relation to the patient anatomy at any time after registration, you must re-register.

Follow these steps to touch the landmarks on the patient:

1. Place the tip of the registration probe that you previously verified on the patient at a landmark. Do not indent the skin. You can touch the landmarks in any order.
2. Depending on your surgeon preferences, do one of the following to store a point:
 - Hold the registration probe steady and continue to hold the probe for 2 seconds.
 - Hold the registration probe steady and press the footswitch.
3. Depending on your surgeon preferences, do one of the following to move to store the next point:
 - If you are using the **Stationary Probe** registration preference, hold the registration probe steady on the point. Once you hear a verification sound or see the landmark circle turn color, move the registration probe to the next point.
 - If you are using the footswitch to store points, release the footswitch before removing the probe from the point.
4. Repeat this process for the remaining landmarks.

When you have touched enough landmarks to compute a registration, the clinical application displays the **Registration Accuracy** value. Lower numbers indicate a more accurate registration.

After a registration is calculated, the touched landmarks change color. The color indicates the distance between the stored landmark and the matching touched point.

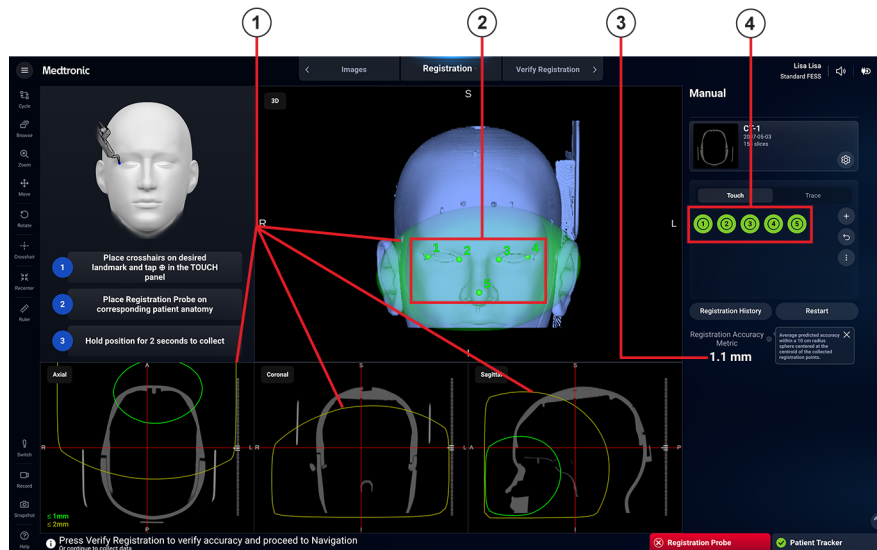
- Green landmarks: Less than or equal to 2.5 mm
- Yellow landmarks: Greater than 2.5 mm up to 5.0 mm
- Red landmarks: Greater than 5.0 mm
- Blue landmarks have not been matched to a touched point.

You can retouch a previously touched landmark by holding the registration instrument's tip on it and holding for 2 seconds or by pressing the footswitch. Retouch previously touched landmarks if you believe that the first touch was incorrect (for example, if your hand or the probe slipped). Additionally, you can clear all registration data for a landmark by selecting the landmark in the landmarks table and tapping **[Clear]**. You can clear all registration data for all landmarks by tapping **[Clear All]**.

If there are multiple red or yellow landmarks, retouch multiple landmarks, including some green landmarks. If the registration accuracy does not improve to your satisfaction, check the placement of stored landmarks in the 2D image views and adjust the placement as necessary.

The clinical application also displays a Zone of Predicted Registration Accuracy as a green shape in the 3D view and as green and yellow outlines in the 2D views.

For more information about the **Registration Accuracy** value and the Zone of Predicted Registration Accuracy, see "Registration accuracy."



- 1 Zone of Predicted Registration Accuracy
- 2 Landmarks that have been touched on the patient anatomy
- 3 Registration Accuracy
- 4 Points touched

5. Tap **[Verify Registration]** to advance to the registration verification task.

If the **Registration Accuracy** exceeds 5.0 mm, the clinical application does not allow you to advance to **Navigation**. In that case, add more landmarks, or if your touch registration contains red and yellow points, retouch multiple landmarks, including some green landmarks. If the registration accuracy does not improve to your satisfaction, consider checking the placement of the stored landmarks in the 2D image views and adjusting as necessary. You can also tap the **Trace** tab and collect more points by tracing on the anatomy, or tap **[Restart]** to start the registration process again. See “Performing trace registration.”

Registration accuracy

When you have collected enough points for the clinical application to compute a registration, the clinical application displays the **Registration Accuracy** value and the Zone of Predicted Registration Accuracy.

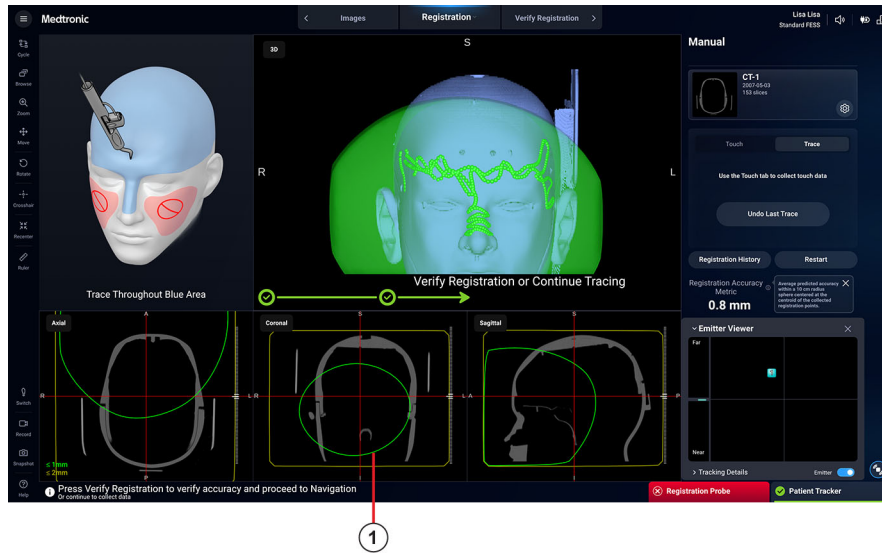
The **Registration Accuracy** value indicates the estimated accuracy due to the registration process only. It is not the overall, system-wide navigation accuracy. Lower numbers indicate a more accurate registration.

Note: Registration Accuracy is a calculation of the average probable error throughout a sphere with a 10 cm radius. The center of the sphere is located at the center of the collected registration points, and that point is typically near the center of the Zone of Predicted Registration Accuracy.

Regions of the image or 3D model within the green Zone of Predicted Registration Accuracy are predicted to have less than 1.0 mm of error due to registration. Regions of the image within the yellow outlines are predicted to have less than 2.0 mm of error due to registration. The clinical application displays this zone in the 3D view and displays cross sections of the zone as green outlines in the 2D views.

As you collect more points during registration, the clinical application will refine the zone. Do not alter your collection or deletion of registration points for small shifts in the **Registration Accuracy** value. Make sure that the registration within the surgical area of interest is satisfactory before navigating.

Note: The Zone of Predicted Registration Accuracy displays only to the edge of the image volume. Large zones or zones near the edge of the image may display flat sides and corners.



- 1 Cross-sections of the Zone of Predicted Registration Accuracy: The area within the green outlines is estimated to have a registration error of 1.0 mm at most.

Verifying registration

Warning: Verify the accuracy of the registration on several known anatomical landmarks before using the registration for navigation. Failure to verify can result in inaccurate navigation.

To verify registration, follow these steps:

1. In the **Registration** task, tap **[Verify Registration]**.
2. Use the registration instrument to touch an identifiable point on the patient's anatomy.
3. Compare the location of this point to the centers of the crosshairs in the images.
4. Repeat the process for at least two more points.
5. Optionally, perform the following tasks to view additional registration information:
6. To view the registration accuracy at a specific location on the image, do one of the following:
 - Pause navigation and place the crosshair at that location to observe the **Registration Accuracy at Crosshair** value that displays on the right side of the screen.
 - Tap the Plans button displayed below the **Registration Accuracy at Crosshair** to open the **Plan** drawer, and then drag the **Traverse** slider to see the **Registration Accuracy at Crosshair** along your plan.

Note: The **Registration Accuracy at Crosshair** will not update while navigating.

7. If you are satisfied with the registration, proceed to **Navigation**.

If the registration is not acceptable, take one of the following actions:

- Return to the **Registration** task and collect more points.
- Tap **[Restart]** to start the registration process again.

Note: **[Restart]** deletes collected points but retains the 3D model and stored landmarks. **[Restart]** also saves your current registration as a previous registration.

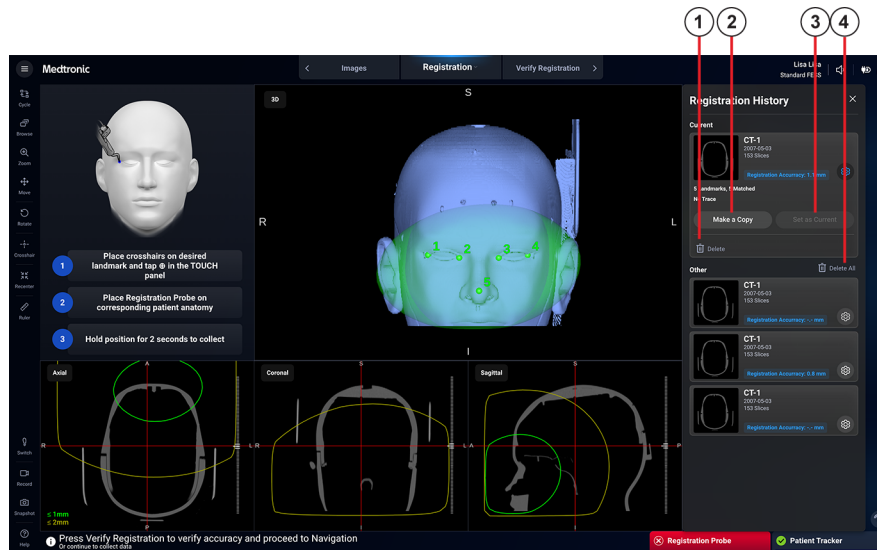
- If you have a previous registration that is accurate, tap **[Previous Registrations]** to open the **Registration History**. Then select the previous registration and tap **[Set As Current]**.

Restoring previous registrations

If you have more than one registration, tap **[Registration History]**. The current registration appears at the top of the list, and older registrations display in the **Other** list in order from the newest to oldest.

- To make a previous registration the current registration, tap the Settings button and then tap **[Set As Current]**. This moves the registration into the Current position at the top of the list. If a registration is deactivated, the **[Set As Current]** button is disabled for that registration. Registrations are deactivated for the following reasons:
 - The patient reference frame has changed since that registration was created.
 - The registration was created in an incompatible procedure.
- To delete a registration in the **Other** list, tap the Settings button on the registration you want to delete, and then tap **[Delete]**. You cannot delete the current registration.
- To delete all registrations in the **Other** list, tap **[Delete All]**, and then tap **[Confirm]**. You cannot delete the current registration.
- To copy a registration, tap the Settings button on the registration you want to copy, and then tap **[Make a Copy]**. The original registration is preserved, and you can modify the copy.

Note: **[Make a Copy]** is disabled if the registration is deactivated.



- 1 **[Delete]**
- 2 **[Make a Copy]**
- 3 **[Set As Current]**
- 4 **[Delete All]**

Navigating

If you plan to use a rotatable blade, see “Using rotatable blades.”

During navigation, the clinical application updates the 2D views in real time to reflect changes in the position of the surgical instrument's tip on the anatomy. The crosshairs in the 2D views represent the position of the instrument's tip. When navigation is active, the crosshairs display green and the instrument displays in the tracking view. When navigation is paused, or if an instrument is not tracking, the crosshairs display red.

You can press the footswitch or tap the Footswitch button on the screen to pause navigation.

Note: You can only use the footswitch to pause navigation when an instrument is tracking in the navigation field.

Warning: The point identified by the crosshairs in the 2D views represents the tip of the instrument.

Warning: Frequently confirm navigation accuracy and system responsiveness during live navigation. Use the navigation instrument to touch several bony anatomical landmarks and confirm that the locations identified on the images match the locations touched on the patient. If accuracy degrades, re-register the patient.

Warning: During navigation, frequently examine instruments and blades for damage. Do not use an instrument that is compromised.

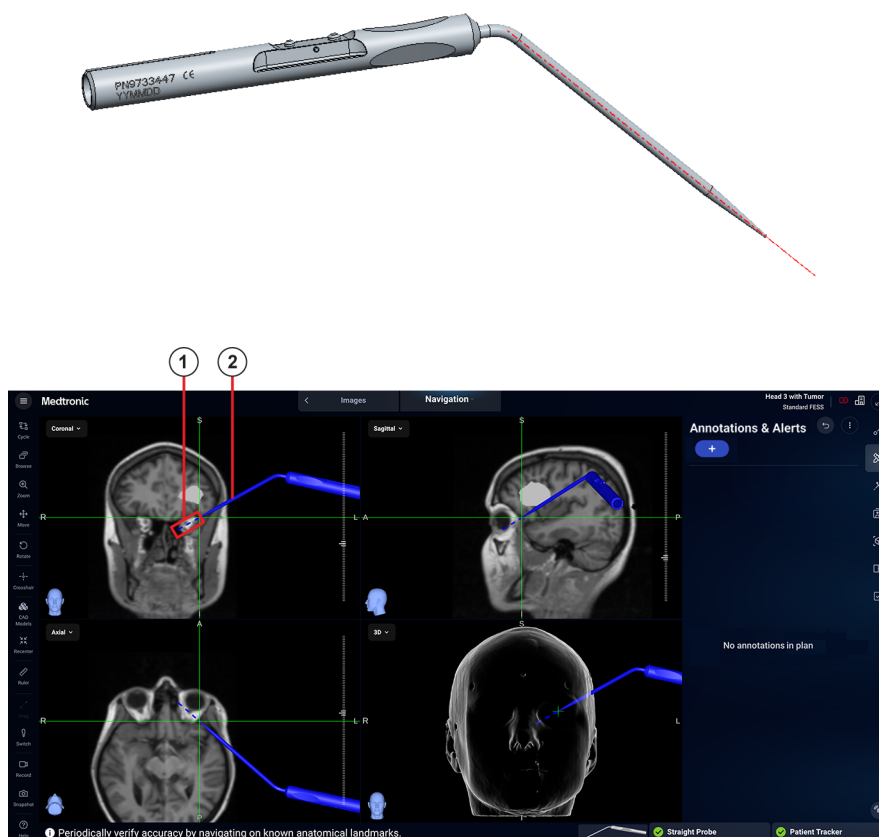
Note: Do not touch electrocautery with metal instruments.

Straight Probe trajectory

When navigating the EM Straight Probe (9733447), the clinical application shows a representation of the probe and a dashed line that shows the present trajectory of the probe.

Warning: The end of the projection is not the same as the tip of the instrument. The projection shows where the instrument will go if you advance it along its current trajectory.

- The dashed line displays for the EM Straight Probe only.
- The trajectory follows the bayonet.



- 1 Representation of the present trajectory of the probe
- 2 Representation of the EM Straight Probe

CAD model overlays

To display CAD models, tap **[CAD Models]** in the left-hand panel.

CAD models are shown for ENT tools that are compatible with the instrument tracker. This does not include debridors and burrs.

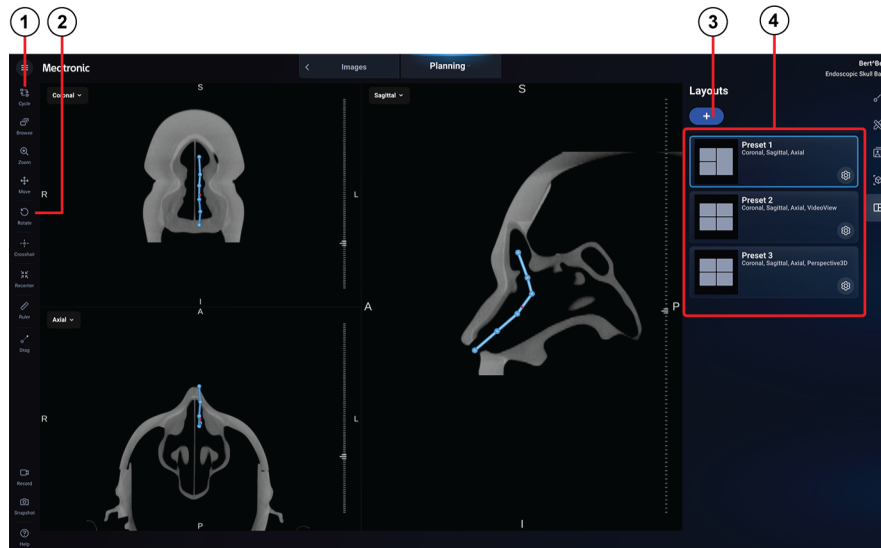
CAD models default to 50% opacity. You can adjust the opacity of CAD models in the **Instruments** drawer.

Adjusting layout configurations

The default image view layouts are based on the selected procedure and task (for example, four quadrants that display the **Axial**, **Sagittal**, **Coronal**, and **3D** views). You can select one of the preconfigured layouts, modify an existing layout and save it, or create your own configuration. Layouts you modify or create are saved to the current surgeon profile and selected procedure.

- To cycle through the saved layouts, tap **[Cycle]**.
- To display a different screen layout, tap one of the layout cards in the **Layouts** drawer.

- To temporarily change a multiple view layout to a one view layout, double-tap the Image Selection drop-down menu in any view. Double-tap again to return to the previous layout.
- To change the display of a view, select the desired view from the Image Selection drop-down menu.
- To save a layout view configuration that can be reused:
 1. Tap the Layouts button.
 2. Tap the Settings button on an existing layout to edit it, or tap **[+]** to create a new layout configuration.
 3. Tap a **Select Layout** card.
 4. To select a different view type, tap the Image Selection drop-down menu in the desired view.
 5. Tap **[Save Layout Changes]**.
 6. (Optional) Name the layout by tapping the name dialog box. Enter the new name and tap **[Save]**.



- 1 [Cycle]
- 2 [Rotate]
- 3 Add a layout configuration
- 4 Preset layouts

Export

Exporting patient data

Follow these instructions to export patient data (including images, merges, and plans), snapshots, or video to media or network locations:

Note: The system is not meant for long-term data storage. Patient data is to be transferred to the system for the patient case.

After case completion, data can be transferred to a PACS or another customer-managed storage device for long-term storage, or the data can be deleted. Data transfer and handling is managed by the customer.

1. Tap the Menu button and tap **Export**.
2. Select one or more patients whose data you want to export.
3. Select an export destination, such as a USB, CD/DVD-ROM, the filesystem, PACS, or another configured Stealth AXIS™ system from the **Destination** drop-down menu.
Select a location for the export, if necessary.
See “Adding a PACS server” to set up a PACS server.
4. Tap the tab for the type of data you want to export.
 - Tap the **All Patient Data** tab to export all data associated with the patient (for example, CT and MR images, snapshots, and videos) and planned data (for example, annotations, plans, and merge data).

Note: This selection archives CT and MR images in the Stealth format. Select the options described in the next step to modify aspects of the exported data.

Note: To view and edit snapshots before exporting, see “Viewing snapshots.”

- Tap the **Snapshots and Videos Only** tab to export snapshots and videos only.
 - Tap the **Patient Coordinate File Only** tab to export only the patient coordinate file.
5. Select the desired export options. The options that appear depend on the data type selection and the data that exists for the selected patients.

DICOM

- If desired, tap the **Description** field and change the text. This text appears in the Patient Name column of the export.
- To anonymize the export, turn on the **Anonymize** toggle and enter prefix text in the **Anonymization Prefix** field. When used, this text replaces the Description text in the Patient Name column of the export and changes patient identifying information in the DICOM tags of the export.

Note: Anonymizing does not delete identifying information in the images themselves, such as user-entered identifying information in any text input fields.

Note: Snapshots are not anonymized.

- Select **Preserve Series Description** to retain the text in the **Description** field in the export. Clear the option to change the Series Description of the export to the entered **Anonymization Prefix**.
- To add an optional accession number attribute to the export, turn on the **Define Accession Number** toggle and enter an accession number in the **Accession Number** field.
- Select **Include Original DICOM Format Images** to add the original DICOM images to the patient data export. When the **Anonymize** toggle is on, the original DICOM image files are anonymized.

File Options

- Select **Include Ultrasound Sequences** to include ultrasound sequences, in mp4 format, in the export.
- If the selected patients have snapshots or videos, they will be included in the export by default, in jpg and mp4 formats respectively. Turn off the **Include Snapshots and Videos** toggle to exclude these files. If you also want to include snapshots and videos in DICOM format, select **Add Snapshots and Videos in DICOM Format**.

Patient Management

Select **Delete Patient After Successful Export** to automatically delete patient data when the export successfully completes.

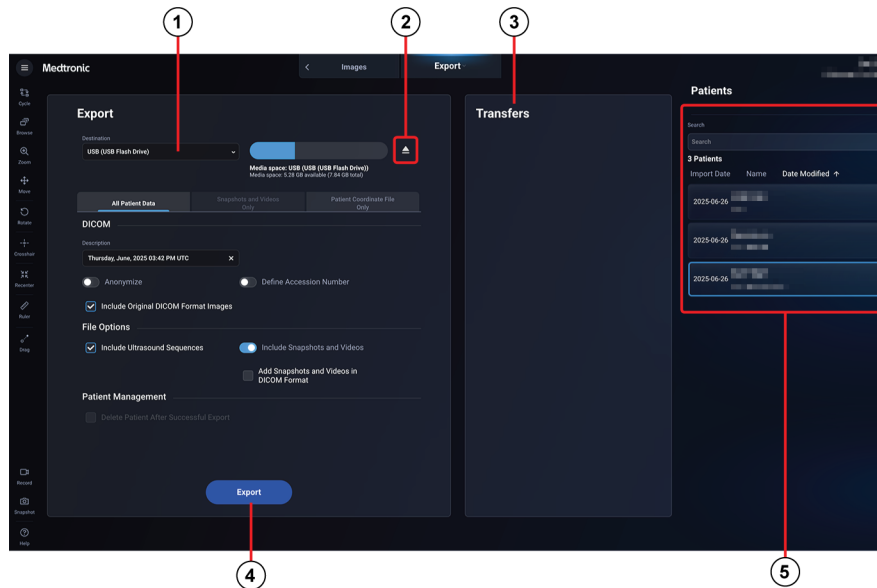
6. Tap **[Export]**.

The **Transfers** list shows export progress.

- Tap **[Cancel]** to stop an export that is in progress.
- Tap **[X]** to close a patient name whose transfer is complete.

Unfinished transfers will be canceled if you navigate away from the **Export** task while an export is in progress.

Note: When you export to a USB drive, the Eject button is displayed on the clinical application screen. After the file transfer is complete, tap the Eject button. If the USB drive has a light, wait for the light to stop flashing before you remove the USB drive from the system.



- 1 Destination drop-down menu
- 2 Eject button
- 3 Transfers
- 4 [Export]
- 5 Patients

Viewing snapshots

Snapshot viewer allows you to view and edit snapshots before exporting them.

1. Tap the Menu button.
2. Tap **[Patient Admin]**.
3. Tap the **Snapshots** tab.
4. Double-tap the snapshot that you want to view or edit. The snapshot enlarges.
 - To rename the snapshot, enter a new name in the **Rename** field, tap **[Rename]**, and then tap **[Confirm]**.
 - To delete the snapshot, tap **[Delete]**.

Managing patients and images

To clone, move, delete, or combine selected patient information, use the **Patient Admin** task. Follow these steps to open the **Patient Admin** task:

1. Tap the Menu button.
2. Tap **[Patient Admin]**.

The **Patient Admin** task displays.

Cloning patients

Cloning a patient includes images, planned data, and any modifications you have made during a session.

Follow these steps to clone selected patients in the **Images** task:

1. Find the desired patient in the **Patients** list.

Note: To search for a patient, enter the patient name or MRN into the text field.
2. If the patient card is not selectable, it is currently in use by someone else. Tap the Clone button.

Note: This is only applicable when using the planning server.
3. If the patient card can be selected, it is not currently in use. Select the patient card to expand it and tap **[Clone Patient]**.

4. On the **Clone Patient** dialog, enter a unique patient name and select the check boxes to include edit history, snapshots, and videos.
5. Tap **[Clone]**.

The new patient card displays in the **Patients** list.

Note: You cannot recombine edits on a cloned patient with the original patient.

Combining patients

Follow these steps to combine two or more selected patients in the **Patient Admin** task:

1. Select patients to combine.
Note: To search for a patient, enter the patient name or MRN into the text field.
2. Tap **[Combine Selected Patients]**.
3. Select the target patient into which the selected patients will be combined.
4. Tap **[Combine]**.

Renaming patients

Follow these steps to rename selected patients in the **Patient Admin** task.

Note: If a patient is used in the current procedure, that patient cannot be renamed.

1. Select a patient.
Note: To search for a patient, enter the patient name or MRN into the text field.
2. Tap the Settings button next to the patient name.
3. In the **Name** field, enter the new patient name.

Deleting a patient

Follow these steps to delete a patient from the system.

Note: If a patient is used in the current procedure, that patient cannot be deleted.

Note: All data associated with the patient, including videos and snapshots, are deleted.

In the **Patient Admin** task:

1. Select a patient.
Note: To search for a patient, enter the patient name or MRN into the text field.
2. Tap the Settings button to expand the card.
3. Tap **[Delete]**.
4. Tap **[Delete]** on the dialog to confirm you want to remove the patient data from the system.

The patient data is removed from the system.

Deleting multiple patients

In the **Patient Admin** task, follow these steps to delete more than one patient:

Note: If a patient is used in the current procedure, that patient cannot be deleted.

Note: All data associated with the patient, including videos and snapshots, are deleted.

1. Tap the **More** button at the top right of the **Patients** list.
2. To delete all patients, tap **[Delete All]**. On the confirmation dialog, tap **[Delete All]**
3. To delete multiple patients, tap the **[Delete Multiple]** button. In the Patients dialog, select the Delete button next to each desired patient. Click **[Done]** when finished.

All patients and patient data are cleared from the system.

Moving images from one patient to another

Follow these steps to move images for selected patients in the **Patient Admin** task:

1. Select patients.
Note: To search for a patient, enter the patient name or MRN into the text field.
2. In the **Images** tab, tap the patient images that you want to move.
3. Tap **[Move Selected Images]**.
 - a. To move the images to a new patient, enter a new patient name and MRN.
 - b. To move the images to an existing patient, select the patient name from the **Patients** drop-down menu.
4. Tap **[Move Selected Images]**.

Deleting images

Follow these steps to delete images for selected patients in the **Patient Admin** task.

Note: Deleting the only image for a patient deletes the patient.

1. Select patients.
Note: To search for a patient, enter the patient name or MRN into the text field.
2. In the **Images** tab, tap the patient images that you want to delete.
3. Tap **[Delete Selected Images]**.
4. Tap **[Delete]** to confirm.
Note: All data associated with the patient, including videos and snapshots, are deleted.

Exiting a case

Follow these instructions to exit a case:

1. Tap the Menu button.
2. Tap the Home button.

The **Select Procedure** screen displays.

Exiting the clinical application

Follow these instructions to exit the clinical application:

1. Tap the Menu button.
2. Tap the Power button.
3. Tap **[Log Out]**.
4. Tap **[Log Out Now]**. Tapping **[Cancel]** will abort the action.

The login screen displays.

Note: If you do not tap **[Log Out Now]**, the system will automatically log out after 25 seconds.

Shutting down the system

Follow these instructions to shut down the system:

Caution: Ensure that the workstation cart is shut down using the clinical application. Shutting down the system by disconnecting from an electrical outlet and letting the batteries drain or holding the power button for more than 6 seconds is not recommended and could damage the workstation cart or put the cart into a condition that requires maintenance.

Once logged into the clinical application, to shut down the system:

1. Tap the Menu button.
2. Tap the Power button.
3. Tap **[Power Down]**.
The clinical application prompts you to confirm.
4. Tap **[Power Down Now]**. Tapping **[Cancel]** will abort the action.
The system will fully shut down.

Note: If you do not tap **[Power Down]**, the system will automatically shut down after 25 seconds.

To power down the system from the login screen:

1. Tap the Power button in the lower-right corner to access the drop-down menu.

Note: The Power button does not display during remote planning sessions.

2. Tap **[Power Down]**.
3. The system prompts you to confirm.
4. Tap **[Power Down Now]**. Tapping **[Cancel]** will abort the action.
5. The system will fully shut down.

Note: If you do not tap **[Power Down]**, the system will automatically shut down after 25 seconds.

Restarting the system

Follow these instructions to restart the system.

1. Navigate to the login screen by following the instructions in “Exiting the clinical application.”
2. Tap the Power button in the lower-right corner to access the drop-down menu.

Note: The Power button does not display during remote planning sessions.

3. Tap **[Restart]**.
The system prompts you to confirm.
4. Tap **[Restart Now]**. Tapping **[Cancel]** will abort the action.

Note: If you do not tap **[Restart Now]**, the system will automatically restart after 25 seconds.

Troubleshooting

Image import problems

Find the symptom in the table below and follow the troubleshooting steps:

Symptom	Troubleshooting steps
You imported a series and slices are missing in the area of anatomical interest.	• If you imported the series from a PACS, ask Radiology to make sure that all slices are present in the original scan. • Ensure that slice spacing and thickness remain the same throughout the scan. Sometimes gaps occur because the spacing and thickness changes in the scan based on how radiology acquires or processes the scan. • If you imported the series from media, use a DICOM viewer to make sure that all slices are present in the original scan. • If all slices are in the original scan, delete the imported series and reimport it. To delete the series, see “Deleting images.” • If the problem persists after you reimport the series, contact Medtronic technical services.
The images do not contain the entire anatomical region of interest needed for the procedure.	Adjust the scanner or the patient so that as much as possible of the region of interest is in the scanner's field of view and then rescan the patient.
You need to correct the orientation of a series due to mislabeling in the scanner.	Tap [Correct] , then tap the options to correct the orientation.
The level and width adjustments seem too sensitive (for example, they change from very light to very dark, and it may be difficult to show defined anatomy).	1. In the Images task, tap the arrow under Series to display details about the patient series, including the original bit depth. 2. In the Patient Images list, tap the Settings button on the patient series. 3. Tap the 2D tab. 4. Tap the Bit Depth drop-down menu and select a bit depth that is lower than the original. 5. Adjust the Level and Width sliders.
You need to move a series from one patient name to another.	See “Moving images from one patient to another” for details.
You need to change the modality of a series.	Select the patient series in the Patient Images list, tap the Settings button, and select the correct modality.

Symptom	Troubleshooting steps
A warning message that includes: "Minimum of three slices required."	Ask radiology to review the scan to comply with the Imaging Protocol.
A warning message that includes: "This exam is restricted from use in the tasks below: Registration."	If you want to include the series in image views for planning or navigation, you can manually merge the series with a series that is acceptable for registration.
A warning message that includes: "This may affect functionality or navigation accuracy of the system."	Ask radiology to reformat the scan to comply with the Imaging Protocol.
When you try to search for images on a PACS, a warning message displays.	After each of the following steps, try your search again to determine whether the step resolved the issue: 1. Make sure the network cable is plugged in on both ends if using a wired configuration. 2. If using wireless connectivity, ensure that an adequate signal is available. 3. See "Running DICOM tests." 4. Check the network settings in the Admin application: a. Tap the Menu button. b. Tap [Admin]. 5. Make sure the ethernet port at the facility has an active network connection. 6. Make sure there are no other devices on the facility's network that are using the same IP address and network configuration. 7. If using wireless connectivity, ensure that adequate signal is available. If the problem persists, contact Medtronic technical services.

Merge tips

When you merge images, the merged set is always used in the **Planning** and **Navigation** tasks.

Goal	Steps
To view only one of the merged images in the Planning or Navigation tasks	Tap [Images] , then tap the Show button on all the images that you do not want to view. When the Show button has a line through it, that image is not visible.
To remove an image from the merged set	Go to the Images task and turn off the Select toggle on the images that you want to remove. Note: If you deselect the image that was previously the reference image, your planning data associated with that image will be removed.

Planning and Navigation tips

Find the goal in the table below and follow the steps:

Goal	Steps
To display dots instead of crosshairs in 2D views	The following methods display crosshairs in 2D views: • Update preferences 1. Tap the Menu button, and then tap [Preferences]. 2. Select the procedure. 3. Tap the Views tab. 4. Find Crosshair Visibility and tap Dot. • Tap [Crosshairs], and then tap [Dot]. • Press <F4> on the keyboard to switch between crosshairs and dots.
To find an existing annotation in the 2D or 3D image views	1. Open the Annotations and Alerts drawer and make sure that the annotation is set to visible by tapping the Show button. 2. Tap the annotation you want to see. The crosshairs update to that location. Note: Annotations are not visible in Guidance view.

Goal	Steps
To change an annotation's name or color, or to delete an annotation when those options are not active	1. Tap the Lock button to unlock the annotation. 2. To delete a single annotation, do one of the following: Tap the Settings button on the annotation and tap [Delete]. Swipe an annotation to the right and then tap the Delete button.
To update the location of an annotation	Move the crosshairs where you want to place the new annotation and tap [Set Location] .

Registration troubleshooting

Find the symptom in the table below and follow the troubleshooting steps:

Note: LSB procedures allow touch registration only. For these procedures, omit troubleshooting steps that include trace registration.

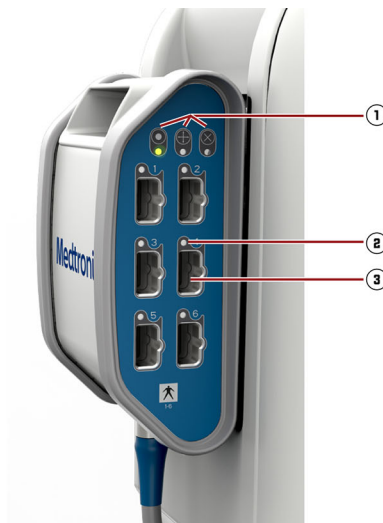
Symptom	Troubleshooting steps
An error message displays, "The current instrument cannot be used for registration."	Use the EM ENT Registration Probe.
• Registration accuracy is not sufficient or an error message displays, "To proceed to Navigation, the overall registration accuracy must be below 5.0 mm. Continue to collect points." • An error message displays, "Continue tracing over a larger area or add touch points." • An error message displays, "Continue to collect data. Registration error is above the recommended threshold of 5.0 mm."	• If fiducials are available, collect them. If trace registration initialization has not been performed, a minimum of 4 fiducials must be acquired. • Touch the locations of all of the stored landmarks. • Store and touch additional landmarks, if possible. • If you are using trace registration, collect more trace points. • If registration still doesn't initialize, or initializes with insufficient accuracy to proceed, restart registration.
Touch initialization has not occurred.	• Touch at least 4 stored fiducials or stored anatomical landmarks. • Touch as many stored fiducials and landmarks as possible. • If registration does not initialize, touch the stored fiducials and landmarks a second time. • Collect a set of trace points that covers a region of the head. • If registration still doesn't initialize, or initializes with insufficient accuracy to proceed, restart registration.
You cannot collect trace points.	• Begin registration more than 1 cm away from patient tracker divot. • Confirm that the patient tracker and registration instrument are actively tracking within the tracking view. • Check the footswitch behavior. To start tracing, either hold the probe steady for 2 seconds or activate the footswitch.

EM connection problems

When you select an EM procedure in any procedure in the ENT clinical application, the EM instrument interface receives power from the system. See the following table for information about each status indicator.

Note: When the EM instrument interface first receives power, some or all of the LEDs may light up briefly.

Interface LED symbols	LED descriptions	Actions to take if the LED is not lit
	Green power LED: Indicates when the system is supplying power to the EM instrument interface.	• Make sure that the cart is powered on. • Make sure that the EM instrument interface cable is connected to the cart. • Make sure that an EM procedure has been selected in the clinical application.
	Green navigation LED: Indicates when the EM system is in the navigation mode.	• Make sure that the EM instrument interface cable is connected to the cart. • Make sure that an EM procedure has been selected in the clinical application. • Make sure that the emitter is turned on in the clinical application. The emitter is turned on by default, but the tracking details area in the clinical application includes a button to turn the emitter off. See "Emitter on/off."
	Yellow fault LED: Indicates, when lit, that the system has encountered a fault. Check the software for more diagnostic information.	If the fault LED is not lit, the system has not encountered a fault. No action is required.



- 1 EM instrument interface status indicator LEDs
- 2 Instrument indicator LED (6 total)
- 3 Instrument port (6 total)

The instrument status LED color relays real-time information about EM instruments and the EM system controller. When you plug in an EM instrument, the LED is orange until the system is ready to track the instrument. When the system is ready to track the instrument, the LED turns green.

Instrument indicator LED color	Actions to take
Green - Indicates that the instrument is properly connected and can be tracked by the system.	No action needed.
Orange - Indicates that the system has detected the instrument, but the system is not ready to track the instrument. When the system is ready, the indicator turns green.	Instrument initialization can take several seconds. If the problem persists, plug the instrument in to a different port, or plug in a different instrument.
Not lit - Indicates that no instrument has been detected. Either no instrument is plugged in, or a failure has occurred.	Plug in an instrument, or if an instrument is already plugged in, make sure that the instrument connector is properly seated in the port. If the problem persists, plug the instrument in to a different port, or plug in a different instrument.

Troubleshooting instruments

Status messages about instruments may appear either on the bottom-left portion of the screen or in the tracking details.

To view tracking details in the **Instruments**, **Registration**, or **Navigation** task, tap on an instrument name and the tracking view opens. Tap **[Open Tracking Details]**.

Check the instrument's color status and status message, and follow the troubleshooting steps below:

Note: If the troubleshooting steps do not resolve the issue or if the issue persists, call Medtronic technical services.

Instrument status or error message	Troubleshooting steps
The instrument displays red status in the tracking view without a geometry error metric.	• Make sure that the tracked part of the instrument is no more than 39.0 cm (15.4 in) from the face of the side emitter or no more than 42.5 cm (16.7 in) from the face of the flat emitter. • Make sure that the tracker is at least 8 cm (3 in) from the side emitter or 5 cm (2 in) away from the flat emitter. • Move any metal objects (such as the endoscope's camera) farther away from the instrument. • Try a different port on the EM instrument interface. • Turn the emitter off and on in the clinical application. • Make sure that the Emitter toggle is turned on in the clinical application.
The instrument displays orange status in the tracking view.	• The instrument is initializing with the clinical application. Allow at least one minute for the clinical application to establish communication with the tracking system and emitter. • Try plugging the instrument into a different port on the EM instrument interface.
The instrument displays red status in the tracking view with a high geometry error metric.	• Make sure that the tracked part of the instrument is no more than 39.0 cm (15.4 in) from the face of the side emitter or no more than 42.5 cm (16.7 in) from the face of the flat emitter. • Move any devices that emit electromagnetic noise (such as electric drills and electrocautery devices) away from the instrument. • Move any metal objects (such as the endoscope's camera) farther away from the instrument.
A banner message displays "Localizer disconnected." The emitter or EM instrument interface also displays red status.	• The localizer has not established communication with the system. Make sure that the emitter and EM instrument interface cables are connected to the system. • Disconnect and reconnect the emitter and EM instrument interface cables to the system.
The clinical application displays a "faulted" error message next to the instrument name, emitter, or EM instrument interface.	• Disconnect and reconnect the emitter and EM instrument interface cables to the system. • The emitter, EM instrument interface, or instrument has encountered a fault.
The emitter or the EM instrument interface display orange status.	• The devices are initializing with the clinical application. Allow at least one minute for the clinical application to establish communication with the tracking system and emitter. • Disconnect and reconnect the emitter and EM instrument interface cables to the system. • Restart the clinical application.
The emitter or the instruments display gray status.	• The EM system is not navigating. Make sure that the emitter and the EM instrument interface are connected to the system. • Make sure that the Emitter toggle is turned on in the clinical application.
A banner message displays, "Instrument not connected."	Plug an EM instrument into the EM instrument interface.
"Current instrument cannot be used for registration. Use a valid probe for registration."	Use the EM ENT Registration Probe.

Instrument status or error message	Troubleshooting steps
An instrument is plugged into a port and it does not display as red or green in the tracking details.	• Unplug and replug the instrument. If the problem persists, try a new instrument. • Try plugging the instrument into a different port on the EM instrument interface.
All instruments display red status.	Check for interference near the emitter. Make sure that no cables are draped over the emitter.

The following troubleshooting information applies to reusable instruments with EM ENT instrument trackers attached.

Additional instrument problems	Troubleshooting steps
The tracked instrument image and name display incorrectly, and navigation is noticeably inaccurate. The crosshairs are at least 2 cm away from the instrument's tip.	Verify the instrument to which the EM ENT tracker is attached and confirm navigational accuracy. Warning: If you transfer a tracker from one instrument to another, you must verify the instrument that now has the tracker. For example, if you move the instrument tracker from the registration probe to the straight suction and do not verify the straight suction, the clinical application still shows the tracked instrument to be the registration probe. In addition, navigation is noticeably inaccurate.
The clinical application makes a negative sound (bonk) and displays the distance from the instrument's tip to the divot because instrument verification failed.	• Try to verify the instrument again and make sure that the tip of the instrument is in the bottom of the verification divot. • Keep the instrument as perpendicular to the divot as possible. • Make sure that the EM ENT instrument tracker is properly attached. • If the distance to the divot is greater than 2.0 mm or the instrument does not verify, it may be bent or damaged. Try another instrument.

Battery backup

The workstation cart will attempt to run on backup battery power when it is not plugged into an electrical outlet. An audible indicator will sound when the system is running on battery backup, and it will increase in frequency when the battery is low. The battery status icon is displayed in the upper-right corner of the clinical application screen. After a procedure is selected, tap the battery status icon to open the **System Status Menu** and view more information about the battery status. The **System Status Menu** can be accessed during any task except while in the **Procedure Select** task.

One of the following battery icons displays in the upper-right corner of the clinical application screen:

Battery icon	Color	Description
	White	The workstation cart is plugged into an electrical outlet and is charging.
	White	The system's battery backup is in use and has a charge greater than 50%.
	White	The system's backup battery is in use and has a charge greater than 25% and less than 50%.
	White	The system's backup battery is in use and has a charge greater than 10% and less than 25%.
	Flashing orange	The system's battery backup is in use and has a charge less than or equal to 10%.
	Flashing red	The system's battery backup service has detected an error or the workstation cart battery has no charge. Unplugging the system's power cord from the electrical outlet may result in immediate system shutdown.
		Contact Medtronic technical services.

Note: Battery icons do not display in the clinical application on the Planning Station or in a remote planning session.

Accessing remote support

Remote Support is a digital service that enables Medtronic to provide features such as: remote diagnostics, subcomponent level diagnostic alerts, and remote troubleshooting support using a real-time-screen-share feature. Remote Support requires an active, secured Internet connection between the Covered Product and Medtronic for connectivity. The screen-share function requires the person operating the system to initiate a screen-share session by physically opening the Remote Support application with the assistance of Medtronic Technical Support.

If requested to do so by Medtronic technical services, access remote support. For detailed information about accessing a remote support session, see the *StealthAXiS™ IT Guide*.

Note: Remote support is unavailable when using the remote planning feature.

Appendixes

A: Using image merge

The image merge feature is only available when the ENT StealthMerge™ license is installed. Image merge enables you to merge multiple images from the same patient.

Warning: Merging patient images manually may impact the overall system accuracy. Always visually verify the accuracy of each merge before using it for planning and navigation. Failure to confirm accuracy can result in inaccurate navigation.

Warning: Do not use the ENT StealthMerge™ clinical application feature with compact conebeam CT image sets. Compact conebeam CT image sets have not been evaluated for use with the ENT StealthMerge™ clinical application feature.

Warning: In some cases, dissimilar image modalities may not accurately merge with one another. Images with similar anatomical details are likely to merge accurately, whereas images with significantly different details may not merge accurately in every case. For best results:

- When merging more than two images, use the highest contrast image as the reference image. For example, use a pre-operative, high resolution, diagnostic-quality MR-T1 or CT as the reference image.
- Make sure that a significant portion of the skull base is visible in all images so that there are enough anatomical details to complete an accurate merge.

You can merge images in the **Merge Images** task. If you load or acquire more images later in the procedure, you can return to the **Merge Images** task to merge additional images. In the **Images** task, turn on the **Select** toggle in an image series to add the images to the merge set. To remove an image series from the merge set, turn off the **Select** toggle in the image series.

To merge multiple images from the same patient:

1. Tap the **[Merge Images]** task.

The **Merge Images** screen appears.

The clinical application automatically selects a reference image and starts merging the first image in the **Images** list to the reference image. To change the reference image, tap the Settings button on the image series and tap **[Set as Reference]**.

Warning: If you change the reference image after you merge images, the clinical application deletes any existing merges of images that had been automatically or manually merged (such as CT and MR images). You must then re-merge images to the new reference. Changing the reference image does not delete pre-merges or the diffusion merge. Failure to re-merge images to the new reference can result in the inability to plan and navigate.

2. Check the accuracy of each merged exam. To check merge accuracy, tap **Blend** or **Split** and scroll through the slices. Tap the Color button when in the split display to visually enhance features. The reference image will display magenta.
 - If the accuracy of the merge is sufficient for the procedure, tap **[Verify Merge]**.
 - If you want to fine tune the merge, tap **Manual Mode** and follow these steps:
 - a. Tap **Fine Tune**. Directional arrows and rotate buttons appear in the Axial, Coronal, and Sagittal views.
 - b. Tap the Color button when in the split display to visually enhance features.
 - c. Use the directional arrows to move the image 0.1 mm at a time.
 - d. Use the rotate buttons to rotate the image 0.1 degree at a time.
 - e. When the image to merge is aligned with the reference image, tap **[Verify Merge]**.

Note: You can also auto-merge again after manually adjusting. The auto-merge will include the manual adjustments as the new starting position.

- If the merge is not aligned correctly, tap **[Start Auto-Merge]**. Alternatively, you can tap **[Reset Merge]** and turn on **Manual Mode**.
 - a. Drag the image to merge into approximate alignment with the reference image.
 - b. When the image is aligned with the reference image, tap **[Start Auto-Merge]**.
- If an image series does not contain anatomical images (such as a PET or fMRI), the clinical application may automatically associate the series with an anatomical image with the same frame of reference. If the clinical application does not automatically associate the series, follow these steps to link the images:
 - a. Tap the image card for the image that will be the reference image.
 - b. Tap the Link button and select **[Pre-merge]**.
 - c. Tap the image card for the image you want to pre-merge to the reference image.
 - d. Tap **[Pre-merge]**.
 - e. Verify that the match between them looks correct.

- To unlink pre-merged images, tap the Settings button on the desired image and follow one of these steps:
 - Tap **[Unlink Pre-Merge]** on a pre-merge image to break the link to the reference image. The pre-merge image moves back into the **Images** list.
 - Tap **[Unlink All Pre-merged Images]** on the pre-merge reference image to break the link to all pre-merged images. The pre-merge reference image and all other images move back into the **Images** list.
 - 3. Visually confirm the datasets match, then tap **[Verify Merge]**. The next merged image displays for your verification.
- When you merge images, the merged set is always used in the **Planning** and **Navigation** tasks. See “Adjusting views for images and models” if you want to change which images are overlaid together in the display.



- 1 **Fine Tune**
- 2 **[Start Auto-Merge]** switches to **[Stop Auto-Merge]** while merging
- 3 **[Reset Merge]**
- 4 **[Verify Merge]**
- 5 Color button
- 6 Selected images for merge with Link button

B: Editing a 3D registration model

Note: The clinical application builds the registration model automatically, and then you can edit that model.

To edit a 3D registration model, follow these steps:

Note: If you edit the 3D model after you begin a registration, the clinical application saves the 3D model to the previous registration.

1. In the **Registration** task, tap **[Edit Model]**.
The **Model Building** task opens.
2. Drag the **Threshold** slider to adjust the threshold.
3. Tap **[Refine]** to automatically refine the 3D model and create a closed surface for optimal registration.
4. Edit the 3D registration model as necessary. See “Model editing tools.”
5. Tap **[Done]**.

Removing the head holder from a 3D registration model

The clinical application automatically builds a 3D registration model, which may include the head holder. To remove a head holder from the 3D registration model, follow these steps:

1. In the **Registration** task, tap **[Edit Model]**.
The **Model Building** task opens.
2. Use any of the following methods to remove the head holder:
 - To retain only a portion of the 3D model, tap the area you want to retain and then tap **[Retain]**.
 - To delete a portion of the 3D model, tap the area you want to delete and then tap **[Delete]**.
 - To remove the outside edges of the model, swipe a 2D image view into the main window and adjust the cropping square to surround only the portion that you want to keep. The model updates automatically as you adjust the cropping square.
 - To erase part of the 3D model, such as the head holder, tap the **Paint** drop-down button and tap **[Erase]**, swipe a 2D image view into the main window, and then drag on the image to erase the head holder.
 - To delete items that are not connected to the largest part of the model, tap the **Advanced** drop-down button and tap **[Remove Islands]**.
 - To cancel the changes to the model and return to registration, tap **[Cancel]**.

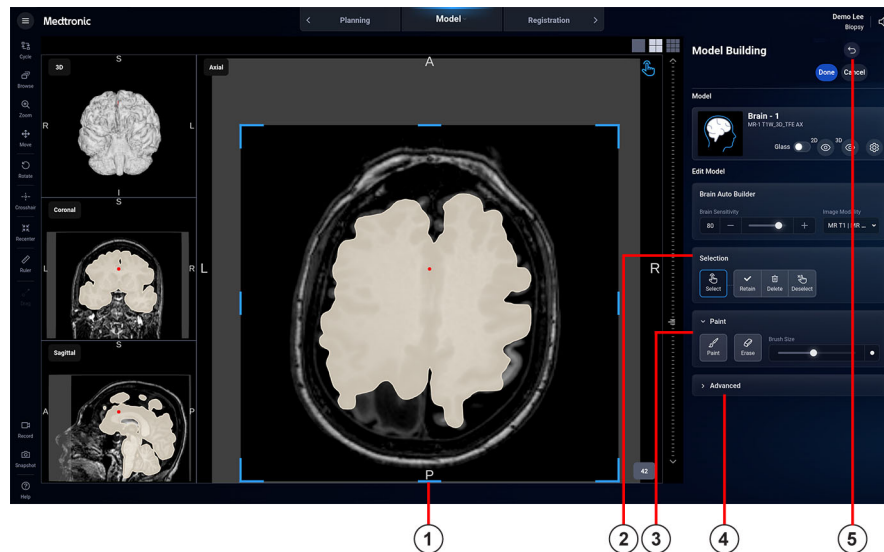
You can further refine the 3D registration model. See “Editing a 3D registration model.”
3. Tap **[Refine]** to automatically refine the 3D model and create a closed surface for optimal registration.
4. Tap **[Done]**.

C: Building and editing models

Model editing tools

When building and editing 3D models, you can use a variety of editing tools for each model.

Editing option	Behavior
Selection card	Displays the buttons for retain, deselect, and delete options. • To retain only a portion of the 3D model, tap the area you want to retain and then tap [Retain]. • To delete a portion of the 3D model, tap the area you want to delete and then tap [Delete]. • To cancel the selection of the 3D model, tap [Deselect].
Paint card	Displays the buttons for painting and erasing within a 2D image. • To use the [Paint] and [Erase] tools, tap [Paint]. • Swipe a 2D image view into the main window, adjust the Brush Size slider, and then drag on the image to paint or erase.
Advanced card	Refines the model with additional options including: • To fill small gaps or to smooth the model, tap the [+] and [-] to adjust the Gap Fill and Smooth sliders. • To reduce the model's size, tap [Contract]. • To increase the model's size, tap [Expand]. • To delete items that are not connected to the model, tap [Remove Islands]. • To fill holes in the model, tap [Fill Holes].
Cropping square	Removes the outside edge of a model. • Swipe a 2D image view into the main window and then adjust the cropping square to surround only the portion you want to keep. The model updates automatically when you adjust the cropping square. • To adjust the cropping square, drag the cropping line to the preferred position. You can adjust the cropping square in any 2D view.
Undo button	Undoes all generic tool actions, all threshold sliders, and Tumor/Other Seed and 3D Paint tools.



- 1 Cropping square
- 2 **Selection** card
- 3 **Paint** card
- 4 **Advanced** card
- 5 Undo button

Building and editing a tumor or other model

Follow these steps to build a tumor model, ventricle model, or any other model type.

1. In the **Planning** task or **Navigation** task, tap the Models button.
2. Tap the Add button.
3. Under **Select Model Type**, tap **[Tumor/Other]**.
4. Select an image.
5. The **Model Building** task opens. Select one of the three methods to build a model: **Seed** (default), **Threshold**, or **3D Paint**.

Note: The clinical application builds the model using only one method. If you select a different method, the current model will be deleted.

The **Seed** tab is the preferred method to build a model. Select seed points and grow the model from those points.

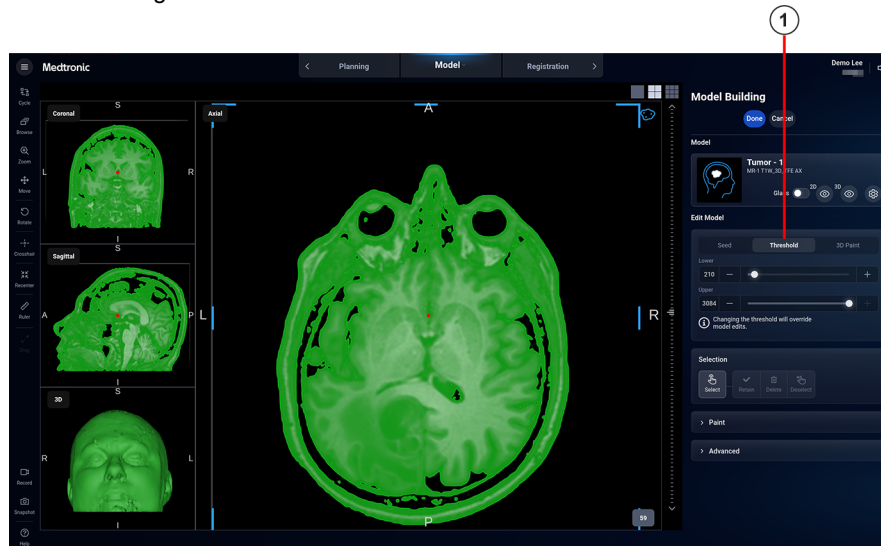
- a. Tap the **[Seed]** button.
- b. Locate a 2D image slice that contains the area of interest.
- c. Tap at least one point in the area of interest.
- d. Locate another slice that contains the area of interest and tap at least one point.
- e. Repeat the steps until you have selected at least three points on different slices.
- f. Tap the Play button to create the model.

Note: You can freeze seed growth by tapping the Pause button. To rewind seed growth, use the **Rewind** slider.



1 Seed tab

Tap the **Threshold** tab to build a model based on light and dark intensity values in the images. Use the **Lower** and **Upper** slider to adjust the thresholding.



1 Threshold tab

Tap the **3D Paint** tab to build a model by painting on multiple image slices that contain the area of interest:

- Tap the **[Paint]** tool.
- Locate the image slice with the smallest portion of the area of interest visible.
- Drag the cursor on the image to paint the area of interest.
- Repeat this process to paint the area of interest on multiple image slices. You must paint in image slices of the same perspective, using only all sagittal, coronal, or axial slices.

Note: Drag on the scroll bar to scroll through the image slices. You can skip some image slices in which the anatomy is similar.

- Tap **[Interpolate]** to create the model.



1 3D Paint tab

6. To display the model in 2D or 3D views in the **Planning** and **Navigation** tasks, tap the 2D or 3D Show button.

Building and editing a brain model

Follow these steps to build and edit a 3D brain model.

1. In the **Planning** task or the **Navigation** task, tap the Models button.
2. Tap the Add button.
3. Under **Select Model Type**, tap **[Brain]**.
4. Select an image.
The **Model Building** task opens, and the clinical application automatically builds a brain model.
5. Edit using brain specific edit options, as appropriate:
 - To change the image modality, select an option from the **Image Modality** drop-down menu.
 - To show more or less brain anatomy in the brain model, drag the **Brain Sensitivity** slider.
6. Continue editing the model with the generic editing tools as necessary.
7. Tap **[Done]**.

Building and editing a bone model

Follow these steps to build and edit a 3D bone model.

1. In the **Planning** task or the **Navigation** task, tap the Models button.
2. Tap the Add button.
3. Under **Select Model Type**, tap **[Bone]**.
4. Select an image.
The **Model Building** task opens, and the clinical application automatically builds the bone model using the **Threshold** method.
5. Use the **Lower** and **Upper** sliders to adjust the threshold.
6. Edit the bone model as necessary.
7. Tap **[Done]**.

Building and editing a skin model

Follow these steps to build and edit a skin model.

1. In the **Planning** task or the **Navigation** task, tap the Models button.
2. Tap the Add button.
3. Under **Select Model Type**, tap **[Skin]**.
4. Select an image.

The **Model Building** task opens, and the clinical application automatically builds a skin model using the **Skin** method.

5. Tap **[Refine]**, and drag the **Threshold** slider to adjust the threshold.
6. Continue editing the model as necessary.
7. Tap **[Done]**.

Building and editing a vessel model

Follow these steps to build and edit a 3D vessel model.

1. In the **Planning** task or the **Navigation** task, tap the Models button.
2. Tap the Add button.
3. Under **Select Model Type**, tap **[Vessel]**.
4. Select an image.

The **Model Building** task opens.

5. Edit using vessel specific edit options, as appropriate:
 - To change the image modality in the **Vessel Auto Builder** method, select an option from the **Image Modality** drop-down menu.
 - Drag the **Threshold** slider to adjust the threshold.
 - To show more or less brain anatomy in the brain model, drag the **Brain Sensitivity** slider.
6. Continue editing the model with the generic editing tools as necessary.
7. Tap **[Done]**.

D: Adjusting views for images and models

Adjusting level and width

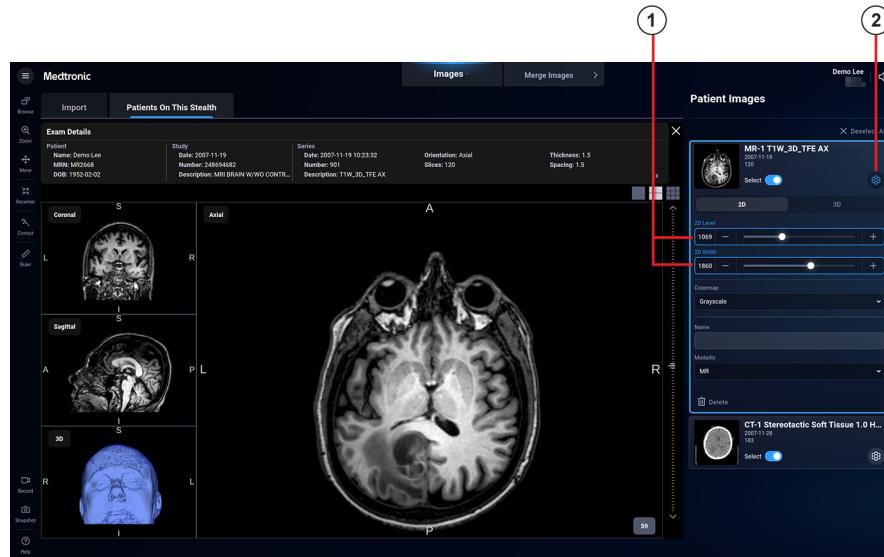
You can use the **Level** and **Width** controls to change how patient images and models display in 2D and 3D views.

Follow these steps to adjust level and width:

1. Tap the Images button to open the **Images** drawer or tap the Models button to open the **Models** drawer.
2. Tap the Settings button on the image or model you want to adjust.
3. Tap the **2D** or **3D** tab.
4. Drag the **Level** and **Width** sliders.

While the **Level** and **Width** controls are selected, you can also adjust level and width by dragging on the views.

- Drag up and down to adjust **Level**.
 - Drag left and right to adjust **Width**.
5. Tap the Settings button or select a different image or model to deselect the level and width controls.



- 1 Level and Width sliders
- 2 Settings button

Displaying multiple images in 2D views

If you merged images, you can display multiple images in 2D views in the **Planning** and **Navigation** tasks. Tap the **Images** button to open the **Images** drawer, and then follow these instructions to adjust how images display:

- To manually display images on top of one another, follow these instructions:
 - Tap the **2D Show** button to display or hide an image.
 - Use the **Opacity** slider to adjust the opacity of an image so that the bottom images can be seen.
- To reorder images, tap on the left edge of the image card and drag up or down in the **Images** drawer.

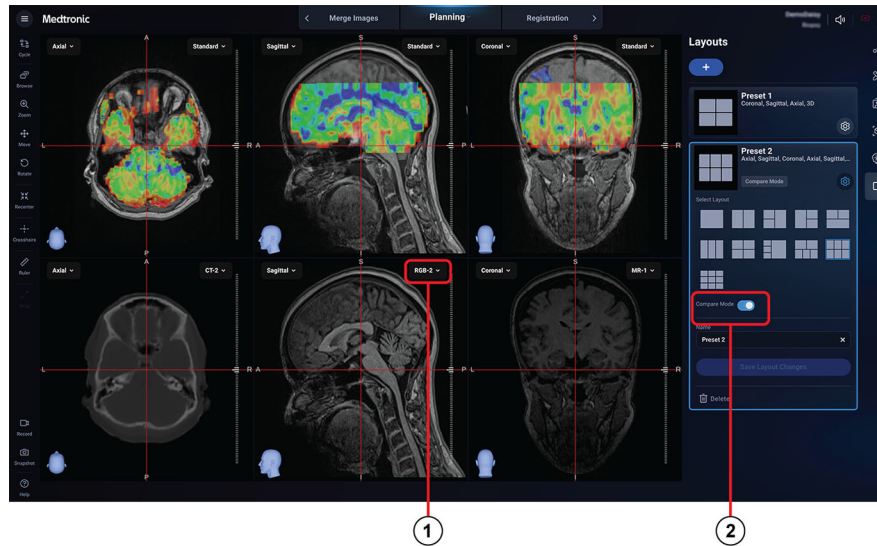
Note: Transparent images placed on top of opaque images will show both images. Images underneath opaque images will not be visible.

Using the compare mode

To compare images, follow these instructions:

1. Tap the **Layouts** button to open the **Layouts** drawer.
2. Tap the **Compare Mode** toggle button on. The view layout changes to a 6-up view with Axial, Sagittal and Coronal views on both the top and bottom rows. If desired, the view layout and view types can be changed.
3. In each view, tap the **Select Image** drop-down menu to change the desired image to display.
4. To save the layout and images selected in each view:
 - Tap in the **Name** field and type a unique preset name.
 - Tap **[Save Layout Changes]** on the compare mode preset card.

The layout is saved for the surgeon profile and procedure in use.



- 1 Select image drop-down menu
- 2 Compare mode toggle

Adjusting colormap or modality of images in 2D views

1. Tap the Images button to open the **Images** drawer.
 2. Tap the Settings button on the image that you want to modify.
 3. To adjust the colormap, open the **Colormap** drop-down menu and then select an option.
 4. To adjust the image modality, tap the Settings button and select a modality from the drop-down list.
- The image modality option is only available in the **Images** task.

Adjusting how models display in 2D views

For a 2D view of a model, you can choose to view contours (outlines) or to fill the outlines.

1. Tap the Models button to open the **Models** drawer.
2. By default, models are displayed as outlines. Tap the **[2D Fill]** toggle to fill the outlines.
The **2D Opacity** slider now affects the opacity of the fill. The outline remains as is.
3. To return to the default view, turn off the **[2D Fill]** toggle.
The **2D Opacity** slider now affects the opacity of the outline.

To reorder models, drag the models up or down in the **Models** drawer. Transparent models placed on top of opaque models will show both models. Models underneath opaque models will not be visible by tapping and dragging left edge of the card.

Adjusting how models and images display in the 3D view

If you want to change how models and images display in the 3D view, follow these instructions.

1. In the **Planning** or **Navigation** task, tap the Models button to open the **Models** drawer.
2. Tap the Settings button on a model or image to open its display options.
3. Adjust how models and images display in the 3D view as necessary, using the following options:

Options for models:

- Turn on **Glass** mode.
- Adjust the **3D Opacity**.
- Choose a **Colormap** from the drop-down menu.
- Update the **Name** field.

Options for images:

- Turn on **Glass** mode.

- Adjust the **3D Opacity**.
 - Adjust the **3D Level** and **3D Width**.
 - Choose a **Colormap** from the drop-down menu.
4. Tap the Settings button or select a different model or image to close the 3D display options.

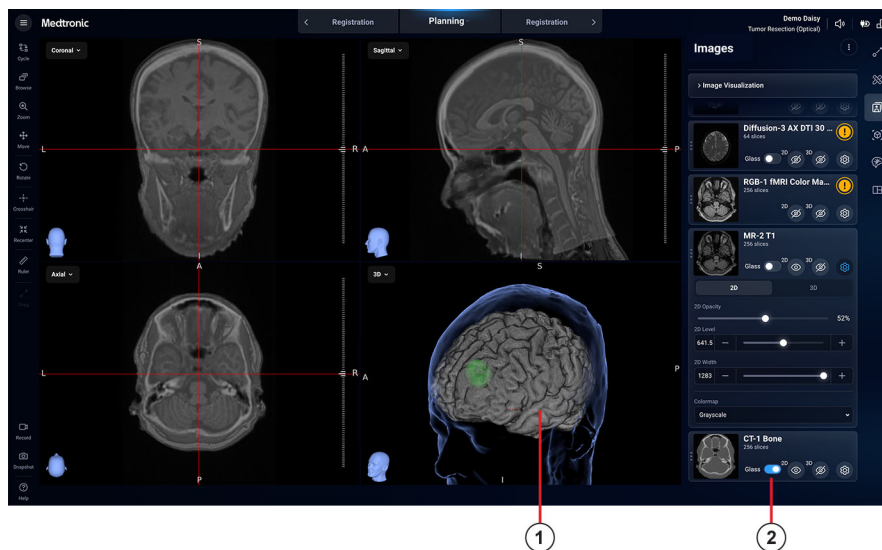
Using Glass mode

Glass mode allows you to see through outer layers of 3D models, such as skin and bone, to increase the visibility of other models, such as tumor models.

To use Glass mode, follow these instructions:

1. In the **Planning** or **Navigation** task, tap the Models button to open the **Models** drawer.
2. Tap the Settings button on a model or image to open its display options.
3. Turn on the **Glass** toggle to enable Glass mode.

In Glass mode, the 3D Opacity is automatically set to 50%. Use the **3D Opacity** slider to adjust the opacity from 0% to 100%.



- 1 Glass mode view
- 2 Glass toggle

Using the 3D Cut mode tool

The 3D Cut mode tool can cut away sections of images and models in the 3D view. To use the 3D Cut mode tool, follow these instructions.

Note: The 3D Cut mode tool does not edit the 3D model. The 3D Cut mode tool only changes the 3D view of the model.

1. In the **Planning** or **Navigation** task, tap the Models button to open the **Images** drawer.
2. Tap **Image Visualization** at the top of the drawer.
3. Turn on the **3D Cut** toggle to enable cut mode.
4. Move the crosshairs to the location you want to see in the 3D view.
The 3D Cut mode tool cuts away a section of the model or image (for which the visibility is currently on) to show the location of the crosshairs in the 3D view.
5. Turn on the **Display 2D Image in Cut Mode** toggle to display the 2D images at the current cut plane.

Adjusting the opacity of images and models

In the **Planning** and **Navigation** tasks, you can adjust the opacity of images and models.

1. Tap the Images button to open the **Images** drawer or tap the Models button to open the **Models** drawer.
2. Tap the image or model that you want to adjust.
3. Use the **2D Opacity** or **3D Opacity** slider to adjust the image or model's opacity from 0 to 100 percent.

Note: In 2D views, you can re-order image panels to place transparent images on top of opaque images.

Note: For a 2D view, if the **2D Fill** toggle is off, the opacity slider affects the outline of the model. If the **2D Fill** toggle is on, the **2D Opacity** slider affects the model's fill, but not the outline.

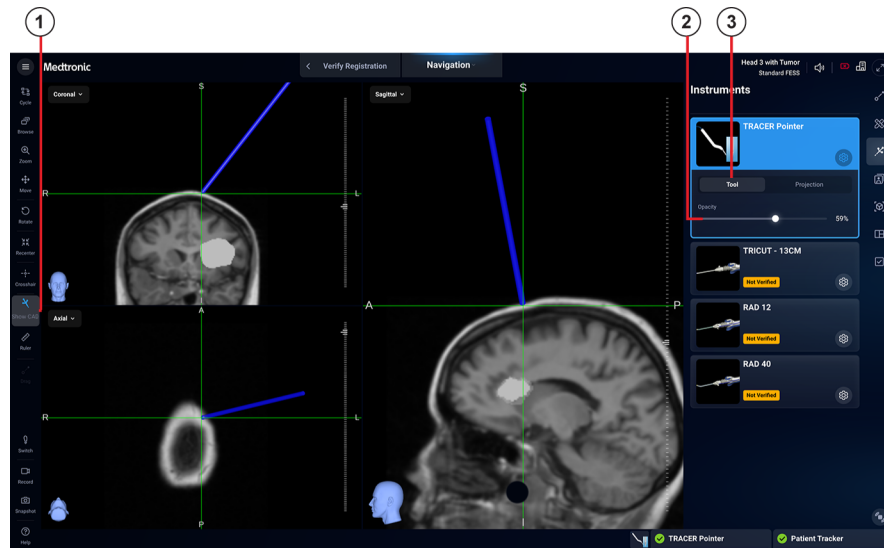


1 2D Opacity slider

Adjusting instrument CAD model opacity

Use the **Opacity** slider to adjust the transparency of the instrument CAD model.

1. In the **Navigation** task, tap **[Instruments]**. To display CAD models, tap **[CAD Models]** in the left-hand panel.
2. Under the desired instrument, tap **[Tool]**.
3. Define the transparency of the instrument CAD model by using the **Opacity** slider.



- 1 CAD Models button slider
- 2 Opacity slider
- 3 [Tool]

E: Creating and editing annotations and alerts

The **Annotations and Alerts** drawer allows you to store points in images and return to them later in the procedure. For example, you can save specific locations along a surgical plan that are defined in 2D and 3D views.

Creating a new annotation

Follow these steps to create a new annotation:

1. In the **Planning** task or **Navigation** task, tap the Annotations and Alerts button to open the **Annotations and Alerts** drawer.
2. Tap on a 2D or 3D patient image to place the crosshairs where you want an annotation.
3. Tap the Annotate button to add the new annotation. An annotation is created at the location of the crosshairs.
If you want to update the location of the annotation, move the crosshairs where you want to place the new annotation and tap **[Set Location]**.
4. Enter a name for the annotation in the **Name** field.
Note: A name will automatically be assigned if it is not entered in the **Name** field.
5. To create another annotation, tap where you want the annotation on the patient image and tap the Annotate button again.

Viewing and hiding annotations

- When you create an annotation, it appears in the **Annotations and Alerts** drawer and the annotation dot is visible in the 2D and 3D image views.
- If you moved away from a specific annotation in the image views and want to return to it, take one of the following actions:
 - Tap the Annotations and Alerts button to open the **Annotations and Alerts** drawer and then tap the specific annotation.
 - Scroll through the image slices in a 2D image view until you locate the slice that contains the specific annotation.
- To view a single annotation's name in the 2D or 3D image views, hold the mouse cursor over the annotation's dot.
- To show or hide a single annotation, tap the Show button.
- To show or hide all annotations, tap the More button and then tap the Show All button.

See "Keyboard shortcuts" for information on displaying all annotation descriptions using a keyboard shortcut.

Creating alerts

An alert can be enabled for an annotation that provides audible and visual feedback when navigating a tool within a user-defined radius. The **Alert** toggle can be turned on for any defined annotations. Follow these steps to create an alert for an existing annotation:

1. In the **Planning** task or **Navigation** task, tap the Annotations and Alerts button to open the **Annotations and Alerts** drawer.
2. Tap the Settings button on the annotation card that you are adding the alert to.
3. Tap the **Alert** toggle to turn on the alert.
4. Set the radius of the alert region using the slider. The alert will add a semi-transparent highlight of this size around the annotation in the same color used for the annotation.
5. To hide and turn off the visual and audible features of an alert, tap the **Alert** toggle. Tap the toggle again to show and turn on the alert.

Editing, locking, and deleting annotations

1. In the **Planning** task or the **Navigation** task, tap the Add Annotations and Alerts button to open the **Annotations and Alerts** drawer.

2. Tap the Settings button of the annotation that you want to modify.

You can modify an annotation in the following ways:

- Edit the location of an annotation.
- Edit the name

Note: You can also edit an annotation name by double-tapping the name in the **Planning** task or the **Navigation** task in the **Annotations and Alerts** drawer.

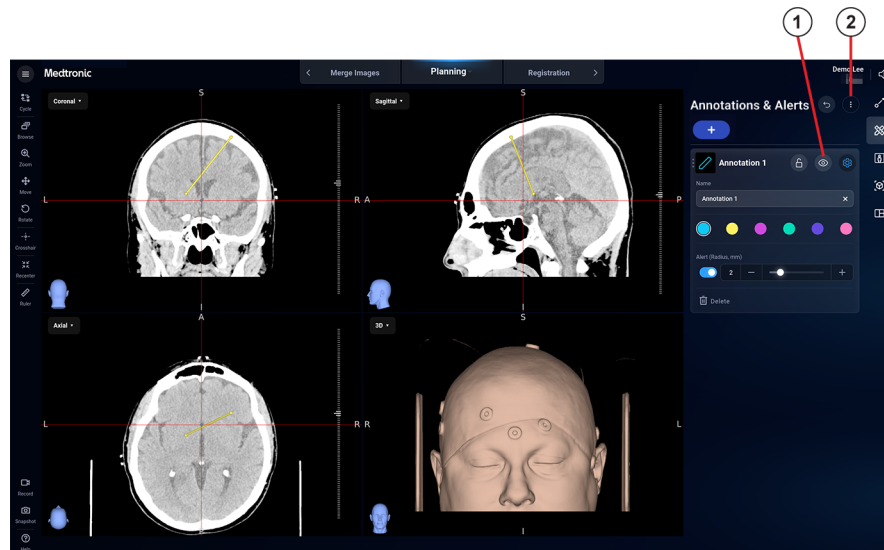
- Change the color
- Lock the annotation to prevent changes
- Delete the annotation

If you made a change to an annotation and you want to undo the change, tap the Undo button.

3. To delete one or more annotations, do one of the following:

Note: You cannot delete a locked annotation. To restore a deleted annotation, tap the Undo button.

- To delete a single annotation, do one of the following:
 - Tap the Settings button on the annotation and tap **[Delete]**.
 - Swipe an annotation to the right and then tap the Delete button.
- To delete multiple annotations, tap the More button then tap **[Delete Multiple]**. Tap the Delete button next to each annotation you want to remove.
- To delete all annotations, tap the More button then tap **[Delete All]**.



- 1 Show button to show or hide annotations
- 2 The More button to **[Hide All]**, **[Delete Multiple]**, or **[Delete All]**.

F: Creating accuracy checkpoints (optional)

Creating five fixed, identifiable points on the anatomy in the sterile field enables you to realign the current registration if you bump or otherwise move the patient reference frame in relation to the patient anatomy. Create the checkpoints immediately after a successful registration.

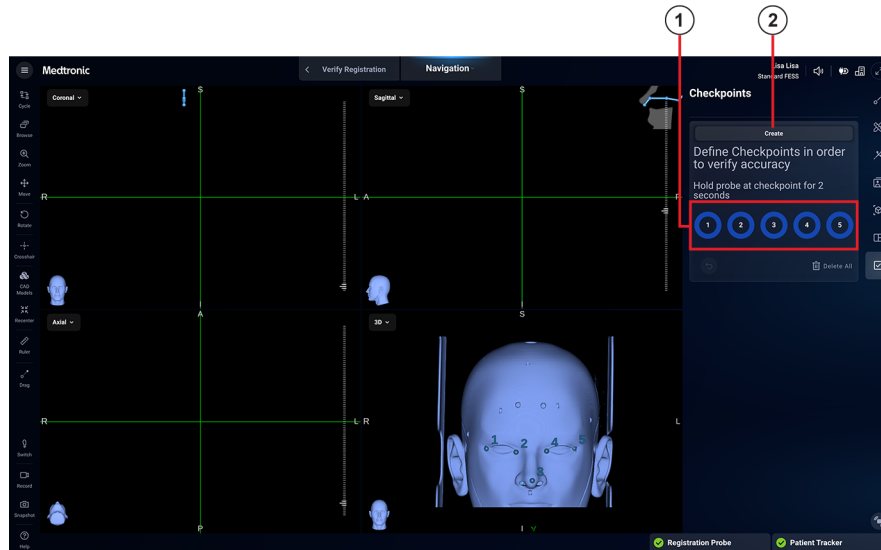
Note: Realigning checkpoints only applies to lateral skull base procedures.

Warning: If you do not create accuracy checkpoints and the patient reference frame moves relative to the patient anatomy at any time after surgery begins, you must re-register the patient or abandon use of the system.

Follow these steps to create accuracy checkpoints in the **Navigation** task.

1. Tap **[Checkpoints]** to open the **Checkpoints** control panel.
2. Place the sterile registration instrument on the first point and hold it steady for 2 seconds or press the footswitch.
The system stores the point in the **Checkpoints** table, makes a confirmation sound (chime), and advances to the next point.
3. Repeat the steps to store 4 more accuracy checkpoints.

If you need to delete an accuracy checkpoint, tap the Undo button or tap **[Delete All]**.



1 Checkpoints table

2 Create tab

Realigning to accuracy checkpoints (optional)

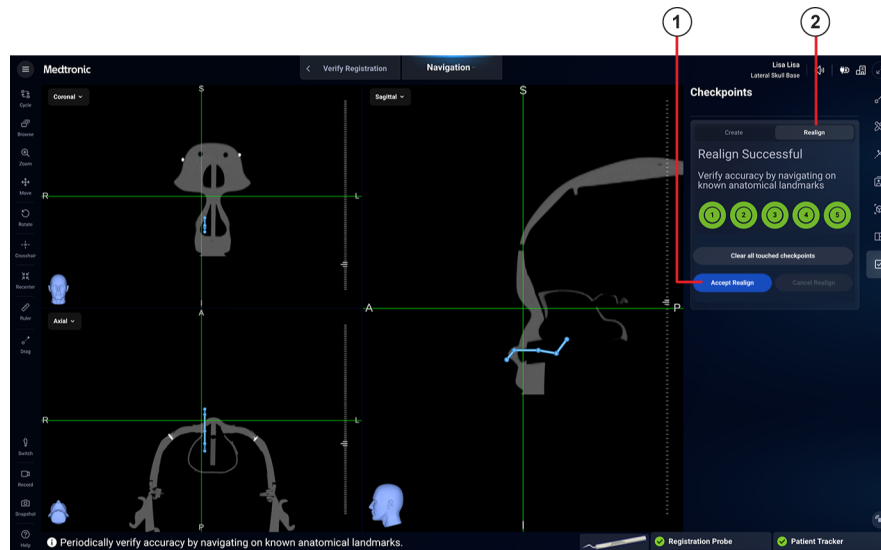
If you created accuracy checkpoints, you can use them in the **Navigation** task to confirm registration accuracy or to realign the registration.

- At any time during the procedure, touch a navigation instrument to any of the accuracy checkpoints and observe the **Distance to Checkpoint** metric in the clinical application banner message.
- A distance in excess of 2 mm may indicate that the patient reference frame moved in relation to the anatomy. If the distance is greater than 2 mm, check navigational accuracy by touching several anatomical points.

To realign the registration to the accuracy checkpoints, follow these steps:

Note: Realigning checkpoints only applies to lateral skull base procedures.

1. In the **Checkpoints** control panel, tap the **Realign** tab.
2. Tap a stored accuracy checkpoint in the **Checkpoints** table.
3. Place the tip of a registration probe on the patient at the first checkpoint. Do not indent the skin.
4. Press the footswitch or hold the probe steady for 2 seconds.
The system stores the location of the checkpoint.
5. Repeat this process for the remaining checkpoints.
When you have touched all 5 accuracy checkpoints, the system realigns the registration to the checkpoints.
6. Use the registration instrument to touch identifiable points on the patient's anatomy and compare the locations to the crosshairs in the images.
7. If you are satisfied with the realigned registration accuracy, tap **[Accept]**.
If the accuracy of the realigned registration is not acceptable, tap **[Cancel]** to discard the realigned registration.



1 Accept Realign

2 Realign tab

G: Using rotatable blades

Warning: Only use the rotatable StraightShot™ automated EM tracking blades with the StraightShot™ M-series handpiece.

Warning: View the endoscopic monitor while dissecting. Do not dissect with the StraightShot™ M-series handpiece while actively tracking the instrument in the navigation clinical application.

Warning: Prior to use, examine devices for damage and deterioration. Do not use any bent, damaged, or compromised devices. Abandon use of any device bent or damaged during the procedure. Contact Medtronic technical services.

Warning: The tracker (hub of the rotatable StraightShot™ automated EM tracking blade) is not intended to be submerged in fluids. Submersion will damage it.

1. Install the rotatable blade in the StraightShot™ M-series handpiece and attach the tubing according to the instructions in the Integrated Power Console (IPC™) System User Guide.
2. Connect the rotatable blade cable to any open port on the Stealth AXIS™ system's EM instrument interface.
The clinical application automatically identifies the blade. The rotatable blades are automatically verified EM instruments and do not need to be verified.
3. Upon successful initialization, the blade name and image display in the clinical application. Confirm that the correct blade is displayed.
4. Prior to use, confirm the accuracy of the rotatable blade by lightly touching the blade tip to several bony anatomical landmarks and observing the centers of the crosshairs on the screen.
 - If the accuracy is acceptable, proceed with navigation.
 - If accuracy is still not acceptable with a new blade, check with a different navigated instrument. If accurate with another instrument, try a different blade and ensure proper attachment to the StraightShot™ M-series handpiece. If the accuracy is still not acceptable, re-register the patient.

Warning: Frequently confirm navigation accuracy and system responsiveness during live navigation. If accuracy degrades, re-register the patient.

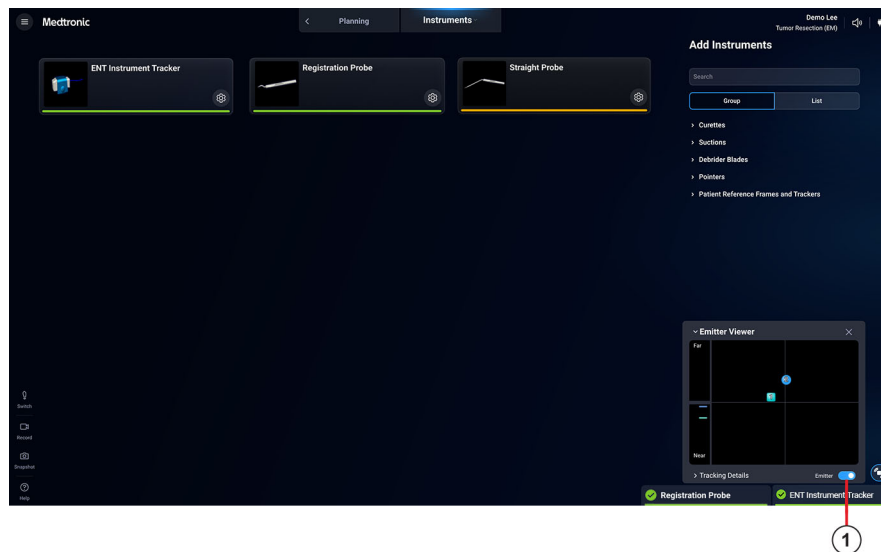
H: Emitter on/off

Turn off the emitter temporarily to allow use of devices that are sensitive to electromagnetic fields. Live navigation is disabled while the emitter is turned off.

Note: When the emitter is unplugged and plugged back in, the emitter will display a connected status even if the **Emitter** toggle is turned off.

1. In the tracking view, tap **[Tracking Details]**.
The status of the emitter displays in the tracking details.
2. To turn off the emitter, tap the **Emitter** toggle.
Note: When the emitter is turned off, live navigation is disabled.
3. To turn on the emitter, tap the **Emitter** toggle again.
The emitter displays **Connected** with a green bar under the emitter.

Note: The emitter takes a few minutes to resume transmitting.



1 Emitter toggle

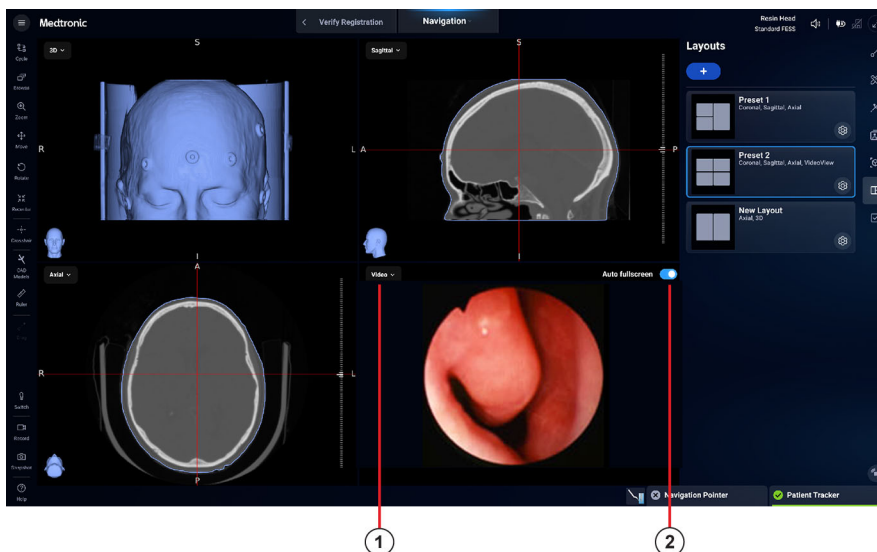
I: Displaying video input

Displaying video input from various devices, such as endoscopes, can be displayed in the clinical application. Make sure that the video input device is properly connected to the system. Confirm that all critical information can be viewed on the source video monitor before importing the video. The system displays video input for secondary display purposes only.

Caution: Do not route endoscope video through other devices prior to routing it to the system. Always connect the endoscope video to the system directly.

- Output the video maintaining a frame rate of at least 4 Hz and a max latency of less than one second.
- Tap the **Image Selection** drop-down menu in an image view and select **Video** to display the video input.
- Tap **Auto full screen** from the video image view to automatically expand the video view to full screen when instruments are not in the emitter field of navigation.

Note: Auto full screen endoscopic video is only available in the **Navigation** task.



- 1 Video
- 2 Auto fullscreen toggle (ENT procedures only)

J: iMRI guidance

Before acquiring an iMRI image, check for any EM patient trackers attached to the patient.

- If the EM tracker is MR conditional, follow the MR conditions in the instructions that accompany the tracker.

Caution: The Non-Invasive Patient Tracker (NIPT) and the Skull-Mounted Patient Tracker (SMPT) are MR Conditional. See the *MRI safety information* instructions for use that accompanies these EM trackers for guidance on MR Conditional use. If the MR system exceeds the MR conditions for these EM trackers, remove the tracker from the patient before acquiring an iMRI image. Failure to follow the instructions can lead to patient harm.
- If the tracker is not MR conditional, remove the tracker from the patient before acquiring an iMRI image.

K: Customizing procedures

You can add customized procedures to a surgeon profile or edit an existing procedure. Customizing procedures allows you to save preferences such as instrument and equipment selection, footswitch behavior, and viewing preferences.

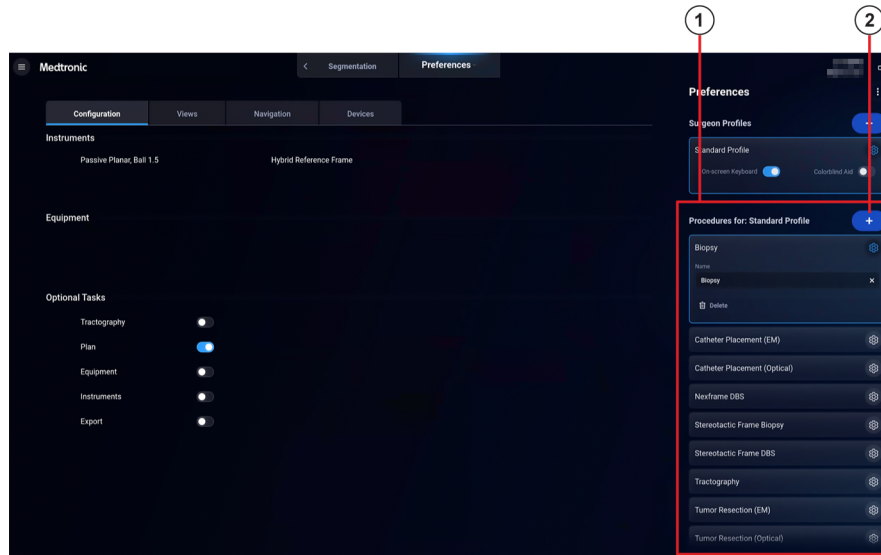
Creating a new procedure

To create a new procedure in your surgeon profile and to customize the procedure selections and surgeon preferences, follow these steps:

Note: To create a new procedure, you must copy an existing procedure or surgeon profile.

1. Tap the Menu button, and then tap **[Preferences]**.
2. Under **Surgeon Profiles**, select the surgeon profile to display a list of available procedures that you can copy.
3. Tap **[+]** next to the procedures list.
4. Tap anywhere in the **Name** field and then enter a name for the procedure.
5. Select a surgeon in the **Copy From** drop-down menu.
6. Select a procedure from the procedure list.
7. Tap **[Create]**.

The procedure is added to the procedure list for the surgeon. All instruments and equipment used in the procedure are automatically saved. You can further edit the procedure to change procedure selections and surgeon preferences. See “Editing a procedure.”



- 1 Procedures list
- 2 [+] (Add New Procedure button)

Editing a procedure

Note: Edits made to surgeon profiles only apply to the system the edits are made on.

You can change the procedure selections and surgeon preferences. To edit surgeon preferences for a procedure, follow these steps:

1. Tap the Menu button, and then tap **[Preferences]**.
2. Under **Surgeon Profiles**, select the surgeon profile to display a list of available procedures that you can edit.
3. Select the procedure that you want to edit.

Note: If you want to make an edit during a procedure, the clinical application will automatically edit the surgeon profile and procedure you are currently in.

4. Edit the selections for the behavior and preferences that you want for each tab.

The settings save automatically as you update them.

Exporting surgeon profiles

You can export surgeon profiles when you set up a new system or if you want to update multiple systems with the same surgeon preferences.

Note: If surgeon profiles are imported between a planning station and a workstation cart, only shared settings will update.

1. Tap the Menu button, and then tap **[Preferences]**.
2. Tap the More button, and then tap **[Export Surgeon Profiles]**.
3. Insert media that you want to export surgeon profiles to, and then select it from the **Media** drop-down menu.

Note: You can export surgeon profiles to a USB, CD, or DVD.

4. Tap **[Export]**.
5. Once the export is complete, tap **[Close]**.

Importing surgeon profiles

You can import surgeon profiles when you set up a new system or if you want to update multiple systems with the same surgeon preferences.

Note: If surgeon profiles are imported between a planning station and a workstation cart, only shared settings will update.

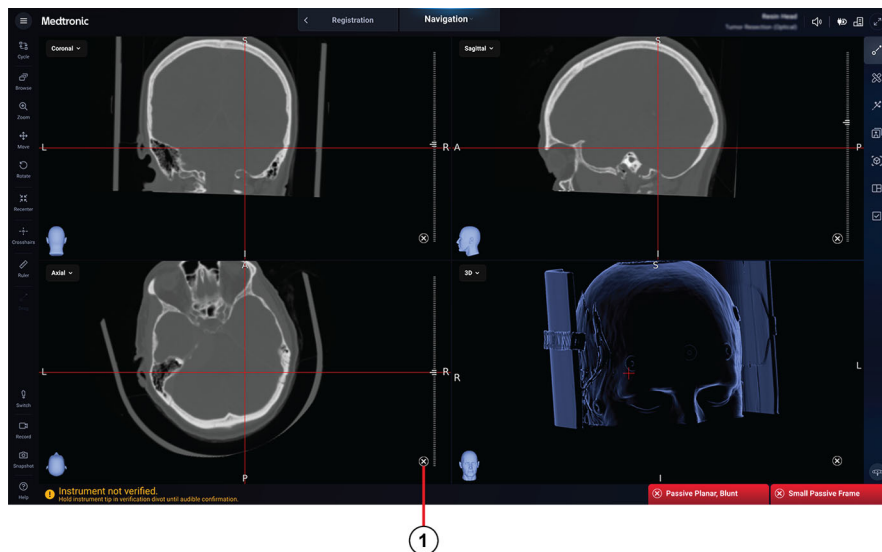
1. Tap the Menu button, and then tap **[Preferences]**.
2. Tap the More button, and then tap **[Import Surgeon Profiles]**.
3. Insert media that contains surgeon profiles you want to import.
Note: You can import surgeon profiles from a USB, CD, or DVD.
4. Select the desired surgeon profile file.
5. Tap **[Import]**.
6. Once the import is complete, tap **[Close]**.

L: Enabling colorblind aid

The colorblind aid feature allows you to use a symbol in the views to show if the system is tracking a navigated instrument.

1. Tap the Menu button, and then tap **[Preferences]**.
2. Under **Surgeon Profiles**, select the surgeon profile that you want to edit.
3. Tap **Colorblind Aid** to switch it to the on position.

When the colorblind aid is enabled for the surgeon profile, the not navigating indicator displays in the bottom right corner of every image view.



- 1 The colorblind indicator that the instrument is not navigating.

M: Controls

Touch and mouse interactions

This table describes touch gestures for using the touchscreen to interact with the clinical application and the equivalent mouse interactions. For touch gestures that require only one finger, you can use a capacitive stylus instead of a finger.

Touch gesture	Gesture description	Equivalent mouse interaction	Clinical application action
Tap	Touch and release the screen with one finger.	Click and release the left mouse button	• Initiate an action with a clinical application button • Select a menu option • Identify a location in an image

Touch gesture	Gesture description	Equivalent mouse interaction	Clinical application action
One finger drag	Place a finger on the screen and move it in a direction while maintaining contact with the screen.	Left click and drag	• Rotate the 3D model • Scroll through menu options • Update a sliding value • Use Paint or Erase in Model Building
Two finger drag	Place two fingers on the screen and move them in the same direction while maintaining contact with the screen.	Click [Move] and then click and drag	Move images
Pinch	Place two fingers on the screen and bring them closer together to zoom out.	Right click and drag down	Zoom out on the images
Stretch	Place two fingers on the screen and move them farther apart to zoom in.	Right click and drag up	Zoom in on the images
Two finger rotate	Place two fingers on the screen and rotate one around the other.	Click [Rotate] and then click and drag	Rotate some 2D views
Flick	Quickly slide and release one finger on the screen.	Left click and drag	Swap images in the Images task
Touch and hold	Touch the screen with one finger and hold it in place.	Left click and hold	To use the virtual [Footswitch] instead of a physical footswitch, touch and hold [Footswitch] until you want to release the footswitch.
Double-tap	Touch the screen twice with one finger in rapid succession.	Double-click	• To temporarily change a multiple view layout to a one view layout, double-tap the Image Selection drop-down menu in any view. Double-tap again to return to the previous layout. • Edit a plan name, plan color, or annotation name

Keyboard shortcuts

The following table contains general clinical application keyboard shortcuts.

Keyboard shortcut	Result
<F1>	Selects view layout preset 1, if stored. At the login screen, switches between small and large text sizes.
<F2>	Selects view layout preset 2, if stored. At the login screen, switches between a brighter or darker display.
<F3>	Selects layout view preset 3, if stored. At the login screen, displays or hides the virtual keyboard.
<F4>	Switches between a dot and a crosshair in the 2D views in the Planning , Navigation , or Tractography task.
<F8>	Displays the crosshair image coordinates panel. The coordinates shown are the current crosshair location. Allows user to specify the crosshair position and crosshair trajectory (Hind).
<F9>	Creates a new annotation when the Annotations interface is open.
<Alt> + <t>: applicable for the workstation and planning carts <Alt> + <t>: (Windows) applicable during a remote planning session <Option> + <t>: (Mac) applicable during a remote planning session	Displays all annotation descriptions and saved measurements.
Left arrow or right arrow	Moves crosshairs by 0.1 mm along the current plan when the Plan drawer is open.

Keyboard shortcut	Result
Up arrow or down arrow	Moves crosshairs by 1.0 mm along the current plan when the Plan drawer is open.

System Status Menu

The **System Status Menu** is in the top-right corner of the clinical application screen during a task on the workstation cart or planning station. Select the status icon area to open the **System Status Menu**.

System Status icons	Description
Volume 	You can control the volume by selecting the Volume button. Note: The volume control is not available in remote planning sessions.
Battery Status 	On the workstation cart only, the battery status icon displays the current detected battery status. See "Battery backup" for more detailed battery status descriptions.
Hospital Network 	The hospital network icon indicates whether you are connected to a network by Ethernet or WiFi, or if you are not connected to a network.

Note: The **System Status Menu** does not display in the **Select Procedure** task.

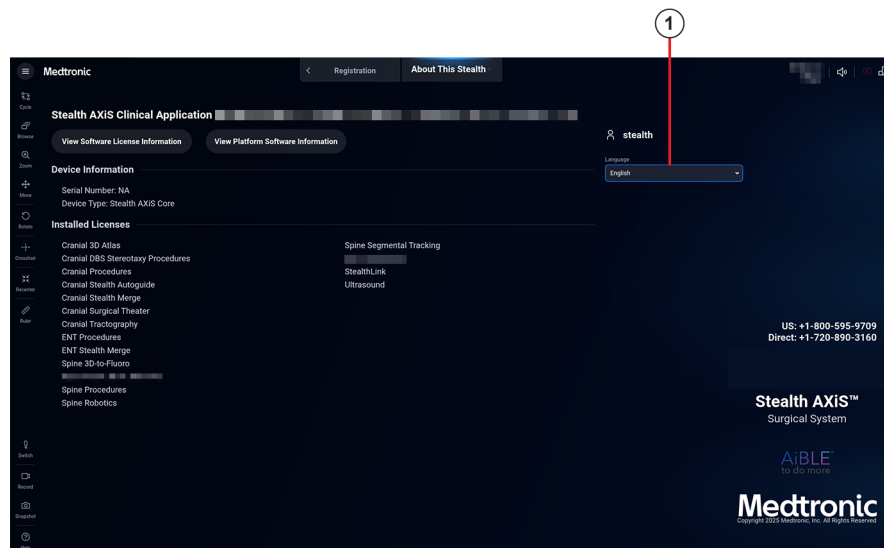
N: Changing the default system setup

Changing the default language

From the **About This Stealth** window you can change the default language.

1. From the main menu, tap **About This Stealth**.
2. To change the default language settings for the application, tap the **Language** drop-down menu and select the preferred language.

Once a language is selected, the application will immediately update to the selected language.



1 Language drop-down menu

O: DICOM Setup

The **DICOM Setup** task contains networking and DICOM settings for the Stealth AXiS™ System. You can add, edit, and delete PACS servers and other Medtronic Navigation™ systems. You can also run configuration tests and configure a system to act as a PACS server.

Use **DICOM Setup** to complete these tasks:

- Add a PACS server
- Edit a PACS server
- Delete a PACS server
- Change the DICOM settings

To open **DICOM Setup**, follow these steps:

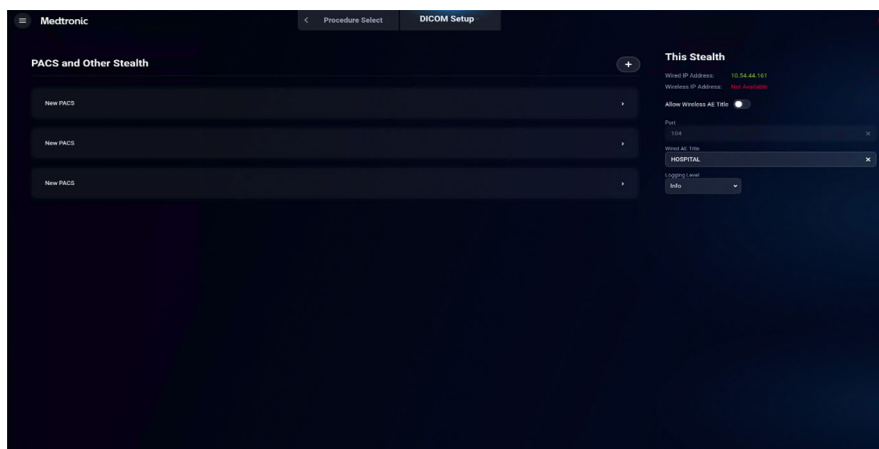
1. Tap the Menu button.
2. Tap **[DICOM Setup]**.

Adding a PACS server

To add a new PACS server, follow these steps:

1. Open the **DICOM Setup** task.
2. Tap **[+]** to add a new PACS server.
3. Enter information for the PACS server:
 - **Name**
 - **AE Title**
 - **IP Address/Hostname**
 - **Port**
4. Select how the system will use the PACS server:
 - **Query**
 - **Storage**
5. To save the PACS server as the default, tap **Default** (optional).

Note: Multiple servers can be configured, but the clinical application can only save one default server at a time.



Editing a PACS server

To edit a PACS server or other Medtronic Navigation™ systems, follow these steps:

1. Open the **DICOM Setup** task.
2. Select an existing DICOM source from the **PACS & Other Stealth** list.
3. You can edit the following:
 - **Name**
 - **AE Title**
 - **IP Address/Hostname**

- **Port**
 - 4. Select how the system will use the DICOM server:
 - **Query**
 - **Storage**
 - 5. Turn on the **Default** toggle (optional) to save the DICOM server as the default.
- Note:** Multiple servers can be configured, but the clinical application can only save one default server at a time.

Deleting a PACS server

To delete a PACS server, follow these steps:

1. Open the **DICOM Setup** task.
2. Select an existing DICOM source from the **PACS & Other Stealth** list.
3. Tap **[Delete]**.
4. Tap **[Delete]** on the dialog to confirm. The server will be removed from the list of available servers.

Running DICOM tests

The following DICOM tests can be run:

- Local Config: Displays information about the local configuration.
- Ping: Sends an ICMP echo to the IP address.
- PACS Port: Attempts to open the specified port.
- Association: Tests that the Stealth AXIS™ system can open a DICOM association to the DICOM server.
- DICOM Echo: Performs the C-ECHO DICOM command.

Running all tests

To run all tests and view results:

1. Tap **[Start Test]**.

The **Test Details** drawer opens. As each test is executed, its corresponding button changes to a status color.

 - Green indicates the test passed.
 - Yellow indicates the test encountered a non-fatal error.
 - Red indicates the test failed with a fatal error.
2. Click **[Details]** to view the detailed output of each test run.

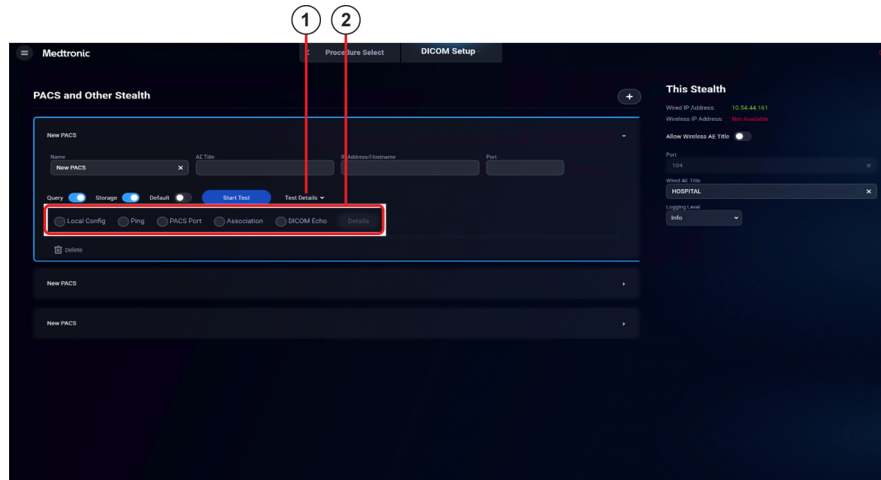
Running selected tests

To select which tests you want to run and display the results:

1. Tap the **All Tests** tab.
2. In the **Test Details** drawer, tap the gray button next to each desired test.
3. Tap **[Start Test]**.

As each test is executed, the corresponding gray button changes to a status color.

 - Green indicates the test passed.
 - Yellow indicates the test encountered a non-fatal error.
 - Red indicates the test failed with a fatal error.
4. Click **[Details]** to view the detailed output of each test run.



- 1 All Tests tab
- 2 Test Details drawer

Changing This Stealth DICOM (PACS) settings

Update **This Stealth** DICOM (PACS) settings to configure the system to act like a PACS server for query, storage, or both.

Only one IP address (wired or wireless) can communicate on the network at any given time. Wireless settings are only shown on systems that have wireless network hardware.

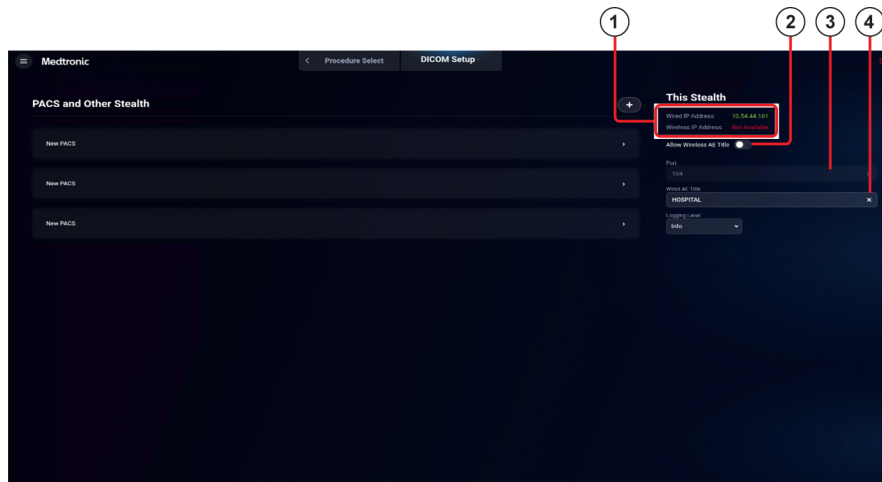
Note: You can use hostnames or IP addresses in the DICOM configuration. These might need to be the fully qualified domain name (FQDN) and not just the hostname by itself, depending on your network setup. Refer to the *Stealth AXIS™ IT Guide* for instructions on how to change IP/hostname.

To enter settings for the system, follow these steps:

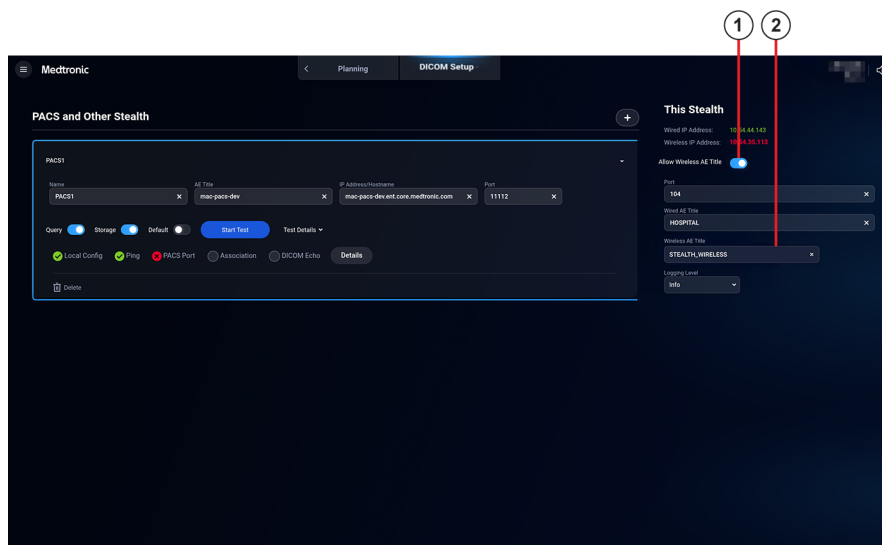
1. Open the **DICOM Setup** task.
2. Change these fields:
 - **Port**
 - **Wired AE Title**

Note: If you are using a workstation cart and the **Allow Wireless AE Title** option is switched on, fields appear for both **Wired AE Title** and **Wireless AE Title**. Both AE title fields can be changed.

Note: Only change the **Logging Level** from its default setting, "Info", at the request of Medtronic technical services.



- 1 If the system is able to communicate on the network, the IP address label is green. If the system is not able to communicate, the label is red. Only one IP address (wired or wireless) can be green at any given time.
- 2 Wireless settings are only shown on systems that have wireless network hardware.
- 3 **Port** field
- 4 **Wired AE Title** field



- 1 **Allow Wireless AE Title** toggle turned on
- 2 **Wireless AE Title** field

P: Buttons

Buttons in Images, Planning, or Navigation

The following buttons display in the **Images**, **Planning**, or **Navigation** tasks.



Cycle



Zoom



Correct



Switch



Browse



Record



Crosshairs



Show CAD



Axial



[Cycle] cycles through the preset image layouts.

[Zoom] magnifies an image. Tap **[Zoom]** and drag up or down on an image.

[Correct] displays options to reorient the images.

[Switch] controls different navigation functions. Tap **[Switch]** as an alternative to pressing the footswitch.

[Browse] scrolls through image slices. Tap this button and then drag up or down on an image.

[Record] captures and records a video of the screen and stores it in the patient data.

[Crosshairs] switches between crosshairs and dots in the 2D views.

[CAD Models] toggles on and off available 3D CAD models over all the views.

The Image Selection drop-down menu allows you to select the content in each view. Tap this button and choose from the drop-down list (for example, **Axial**, **Coronal**, **Sagittal**, **3D**, or **Video**).

The layout buttons change the screen layout to the preset image views.



Move



Recenter



Snapshot



Ruler



Help



Rotate



Drag

[Move] allows you to move an image within a view. Tap **[Move]** and then drag an image.

[Recenter] returns the crosshairs to the centers of the images.

[Snapshot] captures a snapshot of the screen and stores it in the patient data.

[Ruler] displays the distance in millimeters between any two selected points on an image.

[Help] displays relevant help information.

[Rotate] allows you to rotate the 3D model and the trajectory views. Drag the 3D model in the direction that you want it to rotate. In a trajectory view, tap one of the rotation buttons to turn the image 90°.

[Drag] allows you to drag the plan entry or target point. This enables you to update the entry and plan length.

Q: Image label translations

The clinical application displays image orientation labels on the edges of the image views to help identify the patient's orientation. The orientation labels are the initials for the six main anatomical location terms: inferior, superior, anterior, posterior, left, and right.

The image orientation labels may not be translated in the clinical application. They are labeled by the English initials I, S, A, P, L, R, and L.

Label	Meaning
I	Inferior
S	Superior
A	Anterior
P	Posterior
L	Left
R	Right



R: Symbols

Explanation of symbols on package labeling

The following symbols may appear on system equipment, on system packaging, on accessories used with the system, or in this document.



Medical Device



cTUVus Mark certifies the product meets the requirements of a Nationally Recognized Testing Laboratory for compliance with IEC 60601-1 and CAN/CSA-C22.2 No. 60601-1:14.



Federal law (U.S.A.) restricts this device to sale by or on the order of a physician.



Caution



Follow instructions for use.



Consult instructions for use.



Consult electronic instructions for use.



Type BF applied part, in compliance with IEC/UL60601-1.



Type B applied part, in compliance with IEC/UL60601-1.



Single use only. Do not reuse.



Use by date



Batch code



Serial number



Catalog number



Quantity

	Sterile
	Sterilized using ethylene oxide
	Sterilized using irradiation
	Single sterile barrier system
	Single sterile barrier system with protective packaging inside
	Single sterile barrier system with protective packaging outside
	Double sterile barrier system
	Non-sterile
	Do not sterilize.
	Do not resterilize.
	Manufacturer
	Authorized representative in the European Community
	Responsible person in Great Britain (England, Wales, and Scotland)
	Importer
	Date of manufacture
	Keep dry.
	Do not use if package is damaged.
	Fragile, handle with care.
	Keep upright.
	Do not stack.
	Power on. Connect to main power.
	Power off. Disconnect from main power.
	Power on for part of the system.
	Power off for part of the system.
	Power button
	Protective Earth (ground)
	AC Power



Equipotentiality: identifies the terminal that when connected together, bring the various parts of the equipment or system to the same potential, not necessarily being earth potential (for local bonding).



Dead CMOS battery



Do not allow contact with patient. Temperature may exceed limits.



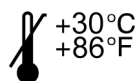
Atmospheric pressure limits



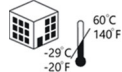
Humidity limits



Temperature limits



Localizer must not be used in ambient temperatures greater than 30°C (86°F).



Storage temperature between -29°C and 60°C (-20°F and 140°F)



Shipping temperature between -30°C and 60°C (-22°F and 140°F)



MR Conditional. Certain accessories that may be used with the system are MR Conditional. See the labeling for those devices for additional instructions on MR conditional use.



Do not disassemble.



USB 2.0 port



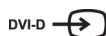
USB port: Connects the system to an external high-speed USB device.



Ethernet port: Connects the system to the site network and medical imaging devices.



DisplayPort for video out



DVI-D port for video in



Footswitch



Camera cart port: Connects the camera cart to the workstation cart.



Robot cart port: Connects the robot cart to the workstation cart.



Workstation cart port: Connects the workstation cart to the robot cart.



Cart docking location



EM instrument interface



EM emitter



Microscope: Connects the system to a surgical microscope. The orange port is for analog microscopes, and the yellow port is for digital microscopes.



SCU ports: Connects an active, wired instrument (patient reference frame or microscope tracker) to the system.



Radio frequency device. Interference may occur in the vicinity of the device.



Protect from heat and radioactive sources

IPX0

Not protected

IPX2

Protected against dripping water when tilted up to 15 degrees

IPX6

Protected against powerful water jets



Locked



Unlocked



Motion lock



Directional lock



Robot arm protective stop



Do not transport the camera cart with a camera or monitor in an undocked position. Always dock the camera and monitor before moving the camera cart. Use the handle to push the cart.



Do not transport the workstation cart with the monitor in an undocked position. Always dock the monitor and close drawers before moving the workstation cart. Use the handle to push the cart.



Do not push the cart from the back of the cart or from the labeled or adjacent surfaces.



LASER radiation emitted from aperture. Do not stare into beam. Class 2 LASER product. Maximum output: <1 mW, CW: 635–670 nm, IEC/EN 60825-1 (2014), ANSI Z136.1 (2014). Complies with 21 CFR 1040.10 and 1040.11 except for conformance with IEC 60825-1 Ed. 3., as described in Laser Notice No. 56, dated May 8, 2019.



Do not dispose of this product in the unsorted municipal waste stream. Dispose of this product according to local regulations. See <http://recycling.medtronic.com> for instructions on proper disposal of this product.



China RoHS compliant. Environmental protection use period of 5 years. Environmental protection use period of 10 years. Environmental protection use period of 50 years.



China RoHS compliant: Electronic Information Products Pollution Control symbol. (The “e” in the logo signifies none of the restricted substances are present above permitted levels.)



Open here

R_x Only

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